

NEURAL BELLMAN OPERATORS

QIAN QI

Peking University

We develop Neural Bellman Operators that construct continuations from their own fitted futures and return feasible economic policies. Exact trained-neural action recovery is connected to a complete policy certificate against the original continuous-action optimum. In the nonlinear investment family, primitive curvature controls the bilinear transition defect and yields convex Bellman values with explicit regularity bounds. Positive interpolation and continuous-law quadrature then give a second-order error account for the actual acquired policy, including action rounding and every remaining date. A prospective comparison covers several state dimensions, longer horizons and complete two-control policies. Neural and conventional proposals receive the same original-optimum targets, direct policy-cost inference and full computational charges. Pure fitted policies and their common-augmented counterparts are reported separately. The original constructive core, broader applications, failed targets and adverse comparisons are retained under their stated hypotheses.

KEYWORDS: Dynamic programming, Neural approximation, Policy certification, Information acquisition, Computational economics.

1. INTRODUCTION

A continuation approximation matters because it changes an economic decision. Its fitting error, the loss of the policy eventually implemented, and the resources needed to verify that policy are different objects. This paper develops Neural Bellman Operators to connect them. A continuation is constructed using its own fitted future, feasible action witnesses are retained through implementation, and a computable policy account determines what has been achieved for the original economic problem.

The first contract is accuracy against the Bellman optimum. Under effective primitive moduli, the constructive backend returns a feasible policy with a prescribed error account. Theorems 2 and 3 establish the construction and its continuous original-owner implementation; the separated resource budget identifies the roles of state coverage, action search and integration. A native envelope and its identical affine-ReLU circuit are two encodings of one construction. This identity is useful for implementation, but does not itself establish an advantage over another numerical method.

The trained backend supplies a second, linked route. Learned nonlinear features propose actions, while verification determines which proposals can be deployed. In the investment model, uniform-innovation integration turns a one-hidden-layer ReLU continuation into a rational piecewise-polynomial action objective. Exact fitted minimization is therefore a finite calculation rather than a second unconverged training problem. The fitted-witness transfer theorem separates center optimization error from acquisition, continuation, continuous-action covering and numerical verification errors. An exact witness removes the first of these errors, not all of them.

We develop two exact implementations of that witness. Lossless interval screening excludes only lattice actions that cannot attain the minimum, and exact comparison preserves the original smallest-index tie rule. Conditional transcript invariance then preserves the acquired policy, its certificates and a prospective controller's stopping decisions. The additional finite-difference construction in Section 10 avoids a favorable-separation premise. On each nonnegative scalar-action piece, the objective is a quartic with a nonnegative fourth-order coefficient. Its second forward difference is nondecreasing. A concave prefix and a convex tail can consequently be searched by rational sign comparisons, including flat objectives, adjacent ties and signed cancellation. No isolated real root or unique surviving action is required.

The action dimension and the state dimension play different roles. The scalar finite-difference result does not become a multivariate theorem by increasing the number of state coordinates. We therefore give a separate exact integration formula for finite mixtures of box-uniform shocks and a terminating rational subdivision result on constrained action lattices. The latter retains the original lexicographic witness, including when an exact fallback is needed. Its worst-case work and the

exponential dependence of the integration formula on the number of independent uniform components are explicit. The two-control experiment implements this extension with a shared capacity constraint and one or two shocks; it is distinct from the complete scalar-action policy services.

The policy account is computed on locally split, non-tensor observation partitions. A one-state residual and value cache encloses a discontinuous incumbent’s continuation. A signed sensitivity calculation for a smooth installed reference retains useful continuation direction without storing a full state-pair table. The nonlinear investment family has a dimension-uniform reference-stability bound, but the work required for a prescribed accuracy still depends on dimension, horizon and the acquired partition. A discontinuous returned actor is handled through every intersected leaf, not by differentiating away its jumps. The separate sample-split approximately-null calculation qualifies inferred nuisance structure and the remaining Bellman comparison under the same maintained-model confidence set.

A prospective service can stop at an economic target before it establishes a small all-state optimality gap. We keep those contracts separate and connect them by a direct Bellman calculation. Whole-state cells and complete continuous-action covers give a lower bound for the original optimum; feasible upper candidates give an implemented policy. Centering both bounds at the same reference value retains pointwise cancellation before a width is formed. A new fixed-policy verification applies that lower bound to the actual policies already returned by the prospective services, without retraining, replacing an action or changing an initial distribution. The convergence argument includes the fixed-cover stability property used by the executable endpoint maps; mere inclusion of a true value is not mistaken for stability under propagated errors.

The empirical analysis distinguishes an implementation comparison from a policy comparison. Matched exact solvers can return the same policy at different computation costs. Different generators can instead return different policies, whose economic ranking requires direct cost evidence. Complete services in dimensions two, four and eight and horizons two, four and six pay for fitting, all attempted searches, whole-cell verification, sequential cost inference and durable return. The current comparison includes common-action, native quadratic, screened ReLU and native

ReLU configurations at the same declared targets. Cold compilation and repeated-use work are reported separately. Earlier tree, compiled-witness, tensor-FVI and adaptive comparisons remain part of the evidence, including their unfavorable outcomes and failed targets.

A broad stress catalogue examines crossing features, cancellation, flatness, exact ties, endpoint resolution and terminating fallbacks. Eight implementations separate several computational mechanisms. A further native two-by-two experiment crosses activation reduction with exhaustive or finite-difference search, holding the programming language and exact objective fixed. Its repeated queries and batch-size interventions isolate effects that a comparison between an unreduced exhaustive implementation and a reduced piecewise implementation cannot isolate. These query experiments do not replace the complete-service clocks. Likewise, fewer exact comparisons do not by themselves imply a lower running time, a cheaper conventional comparison, or a better economic policy.

Related methods. Approximate and fitted policy iteration have established approximation-error analyses (Munos, 2003, Munos and Szepesvári, 2008, Scherrer et al., 2015). Conservative and safe improvement relate a policy update to its incumbent (Kakade and Langford, 2002, Pirotta et al., 2013, Metelli et al., 2021), and tree-based fitted iteration is an important conventional alternative (Ernst et al., 2005). The action-contrast development distinguishes fixed-policy residual transport from state bisimulation metrics, reward transformations and value-equivalent model classes (Ferns, Panangaden, and Precup, 2004, Ng, Harada, and Russell, 1999, Grimm, Barreto, Singh, and Silver, 2020). These are the inherited principles on which the policy account builds.

Interval exclusion and subdivision are standard verification ideas (Moore et al., 2009). Discrete convex analysis supplies a broader language for finite-difference structure (Murota, 2003); the signed neural objective here need not be globally convex. The specific additional result is an exact witness construction for its concave-prefix/convex-tail pieces, with the original tie rule, explicit rational work and a transfer to the same acquired economic policy. The paper’s contribution is the

combined construction–witness–verification–return chain, not a claim that safe improvement, interval pruning or exact arithmetic is intrinsically neural.

The article establishes this chain for finite-horizon expected costs. The complete development article and proof companion retain the recursive-preference, temporal-self, diffusion and game applications under their original hypotheses. All prior theorem statements, original introduction files, experimental records and adverse comparisons remain in the revision and its preservation map. The current introduction organizes that material around the economic contracts being proved and measured; it does not substitute a different model for Neural Bellman Operators.

Original-optimum accuracy of the implemented policy. A feasible witness is a local output; a decision maker ultimately purchases a complete controller. Section 12 supplies a new bridge in the original investment economy. Although its state transition is bilinear, primitive state curvature dominates the possible loss of convexity in the continuation composition. The resulting Bellman functions have explicit Lipschitz and semiconcavity bounds at the original discount. Positive state interpolation and continuous-law quadrature can therefore be budgeted at second order. A separate action-transfer argument certifies the actual barycentric acquired actor rather than only a fitted objective or a nodewise value table.

The prospective comparison in Section 19 applies this contract to pure neural proposals, conventional fitted proposals, and their common-augmented policies across dimensions, horizons and a full two-control economy. Original-optimum accuracy, comparative expected cost and complete release work remain separate endpoints. This completes another part of the original NBO construction without identifying exact action search with a universally superior dynamic-programming method.

Convex dynamic programming and the additional primitive argument. Convex dynamic programs and uniform policy approximation are established subjects. Yang (2020) constructs convex optimization-based Bellman approximations and policy suboptimality bounds for continuous spaces. Our added argument verifies the required regularity for the stated bilinear, capacity-constrained investment primitives, derives coordinate interpolation and continuous-action errors, and carries

these allowances through the recorded acquired actor. It does not claim priority for convex dynamic programming or assume that a signed-weight fitted ReLU is convex. The mathematical object whose curvature is proved is the original Bellman value.

2. THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates $t = 0, \dots, T$ and an invariant compact state domain $X \subset \mathbb{R}^d$. At date t , the action belongs to a nonempty compact set $A_t(x)$. The transition is $F_t(x, a, Z_{t+1})$, the current cost is $c_t(x, a)$, and the terminal cost is $g(x)$. Write $B_0 = 1$ and $B_t = \prod_{j < t} \beta_j$, with $\beta_j > 0$. An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation v , define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \quad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \quad (1)$$

Throughout the deterministic comparison, the evaluations ρ_t are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the experiments ρ_t is conditional expectation, costs are bounded and innovations have their original continuous uniform law. The direct-cost inference in Section 13 is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (1). It forms targets using its *own* fitted f_{t+1} , returns a neural continuation and a feasible policy, and records an economic error allowance. It does not receive the optimal continuation as a training label. Arbitrary trained networks can be evaluated by the first result below. The subsequent constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

2.1. Action-dependent errors and complete-policy loss

Let $H(a)$ be a true current cost and $\hat{H}(a)$ its continuation approximation. If $\hat{a}$ is δ -optimal for $\hat{H}$ on the same feasible action set, then

$$H(\hat{a}) - \inf_a H(a) \leq \delta + \text{osc}_a \{\hat{H}(a) - H(a)\}. \quad (2)$$

Indeed, compare $\hat{a}$ with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 1—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \leq u_t, \quad \mathcal{T}_t^\pi f_{t+1} - f_t \leq v_t, \quad \ell_T \leq f_T - g \leq u_T.$$

Then the actual feasible policy satisfies

$$0 \leq J_0^\pi - V_0^* \leq \sum_{t < T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \quad (3)$$

The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.

PROOF: Backward comparison and cash invariance give

$$V_t^* \geq f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J_t^\pi \leq f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$. The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (3). A shift $f_t \mapsto f_t + c_t^0$ adds $c_t^0 - \beta_t c_{t+1}^0$ to u_t and subtracts it from v_t . The terminal width is unchanged. Q.E.D.

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (3) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

3. CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state–action pairs, state modulus $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and action modulus $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Let

$$d_H(A_t(x), A_t(z)) \leq K_t^A \|x - z\|_1, \quad L_t = A_t^Q + D_t K_t^A, \quad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that $\mathcal{T}_t f_{t+1}$ is L_t -Lipschitz whenever f_{t+1} is L_{t+1} -Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes $x_{t,i}$ with covering radius h_t , form effective feasible k_t -nets. Evaluate their own-future objectives with certified error at most e_t . Minimize the computed labels and retain the corresponding *original* action $a_{t,i}$. A Borel repair $\Gamma_{t,i}(x)$ is feasible at x and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \leq K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \quad (4)$$

The allowance ζ_t concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad i_t(x) = \min \arg \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \quad (5)$$

Return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$. Terminal labels approximate g with error e_T and use the same cone construction with modulus L_g .

THEOREM 2—Own-future construction with feasible witnesses: *The functions in (5) have exact affine–ReLU realizations and Lipschitz moduli L_t independent of label error. The returned policy is feasible and measurable. It satisfies*

$$-e_t \leq f_t - \mathcal{T}_t f_{t+1} \leq D_t k_t + e_t + 2L_t h_t, \quad (6)$$

$$\mathcal{T}_t^\pi f_{t+1} - f_t \leq e_t + D_t \zeta_t,$$

and hence

$$J_0^\pi - V_0^* \leq \sum_{t < T} B_t(D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T(2e_T + 2L_g h_T). \quad (7)$$

With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.

PROOF: The selected node label lies between the node Bellman infimum minus e_t and that infimum plus $D_t k_t + e_t$. The Lipschitz property of the infimum implies that every cone lies above it minus e_t . At a nearest node its cone exceeds the infimum by at most $D_t k_t + e_t + 2L_t h_t$. At the attaining original index, the two feasible pairs and (4) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \leq y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (6). The terminal residual belongs to $[-e_T, e_T + 2L_g h_T]$; Proposition 1 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities $|z| = \text{ReLU}(z) + \text{ReLU}(-z)$ and $\min(u, v) = u - \text{ReLU}(u - v)$ give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. Q.E.D.

The Lipschitz-extension principle is classical (McShane, 1934). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are one method with two encodings. They return the same function and policy. Neither Theorem 2 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

3.1. Original-owner compilation on a tensor cover

On a tensor cover $\mathcal{X} = \prod_{j=1}^d \{x_{j,0}, \dots, x_{j,n_j}\}$, put $S = \prod_j (n_j + 1)$. For arbitrary labels y_i , including inconsistent inexact labels, compute at each grid node v

$$(z_v, o_v) = \min_{\text{lex}, i} (y_i + L\|v - x_i\|_1, i).$$

The index o_v is the original label owner, not the transformed node v .

THEOREM 3—Continuous original-owner identity: *For a closed tensor cell C with corners $\mathcal{V}(C)$, and $x \in C$,*

$$f(x) = \min_{v \in \mathcal{V}(C)} (z_v + L\|x - v\|_1), \quad (8)$$

$$(f(x), i_*(x)) = \min_{\text{lex}, v \in \mathcal{V}(C)} (y_{o_v} + L\|x - x_{o_v}\|_1, o_v).$$

Coordinatewise forward and backward relaxations compute (z, o) in $O(dS)$ additions and comparisons and $O(S)$ storage. After locating the cell, a query uses $O(d2^d)$ arithmetic operations. Exact rational comparisons preserve all ties, including $L = 0$. Using the original owner's repaired action preserves the uncompiled policy, its cost and its certificate exactly.

PROOF: The triangle inequality bounds $f(x)$ above by each corner expression. Choose the lexicographically first attaining site $i_*(x)$. In every coordinate choose the cell endpoint between x_j and $x_{i_*,j}$. At this corner distances add exactly. An owner of strictly smaller transformed value would improve the objective at x ; a tied smaller owner would contradict the original tie rule. Thus this corner has owner $i_*(x)$, proving both identities. Along a coordinate line, forward and backward relaxations respectively minimize over sites on either side, always comparing value–original-index pairs. Iterating over coordinates gives the separable distance transform and its stated work. Rational arithmetic and unchanged repair complete the claim. *Q.E.D.*

The nodal algorithm is the classical distance transform ([Felzenszwalb and Huttenlocher, 2012](#)); preserving the original action through continuous off-grid queries is the policy obligation. If there are $J_{t,i}$ node actions and M_t integration representatives, write $Q_t = M_t \sum_i J_{t,i}$. Literal dense cone evaluation costs $O(\sum_t dS_{t+1} Q_t)$ in

representation work. Compilation replaces this by $O(\sum_t [dS_{t+1} + Q_t d 2^d])$, plus cell location, primitive evaluation, integration, feasible repair, certificate and output. Rational bit lengths and state boxes crossing many cells incur additional costs. There is no dimension-free complexity assertion.

3.2. Acquired states and conventional actors

A b -bit coordinate sensor reports a dyadic cell and its midpoint $\hat{x}$. On the unit cube, $\|x - \hat{x}\|_1 \leq r(b) = d2^{-b-1}$. For the investment capacity $\kappa(x)$ below, clip the original action to the smallest capacity throughout that cell and round down to an action quantum Δ_t . This is feasible at every state in the cell. Relative to an exact true-state repair, its displacement is at most $2K_t^A r(b) + \Delta_t$. Transporting the witness cone from $\hat{x}$ to x adds at most $2L_t r(b)$, plus any certified selector objective error v_t . Thus the implemented-policy account adds

$$\sum_{t < T} B_t \{ (2L_t + 2D_t K_t^A) r(b) + D_t \Delta_t + v_t \}. \quad (9)$$

No continuity of the discrete owner selector is assumed. At uncertain numerical comparisons the simulator retains all possible actors, rather than choosing a favorable one.

For conventional FVI the continuation is multilinear on its realized tensor grid. Its actual modulus is bounded by the largest coordinate edge slope. A nearest-node actor uses a feasible node minimizer and true-state repair. Let h_t be the largest half-cell ℓ^1 radius, L_t^{gr} the Bellman graph modulus, and e_t the node query allowance. Its optimal residual upper bound is $D_t k_t + e_t + L_t^{\text{gr}} h_t$. Its selected-policy upper bound is $e_t + E_t$, where

$$E_t = \sup_x \{ y_{i(x)} + L_t^{\text{gr}} \|x - x_{i(x)}\|_1 - f_t(x) \}.$$

Partitioning cells by nearest-node boundaries makes the expression separately affine in each coordinate. Its maximum is therefore at a partition vertex; the implementation encloses all such vertices. Both uniform and curvature-refined FVI receive this sharper one-sided account, not an artificially doubled baseline. Their acquired-state allowance is derived separately in (61).

4. ALLOCATING RESOURCES TO AN ECONOMIC TOLERANCE

A single resolution parameter imposes an economic restriction that the Bellman problem does not impose. State coverage, action search and integration have different error coefficients and different effects on work. Refining all three at the same rate can therefore spend most of the service budget on the wrong source of error. We separate these choices while retaining the continuation, feasible-witness interpretation and all-state comparison developed above.

4.1. A separated primitive budget

Consider the investment family specified in Section 15.5, with $d \in \{2, 3, 4\}$ state coordinates, discount $\beta = 15/16$, and uniform coordinate grids with N intervals. At a state node use $K + 1$ feasible action fractions. Integrate the common scalar innovation using M equal-probability bins and their midpoints. The three integers need not agree. Let L_t be the primitive graph modulus of the constructed witness continuation, and let D_t be its action modulus. A downward numerical displacement of the action grid is denoted by δ_a .

The state and action covering radii satisfy $h_t = d/(2N)$ and $k_t \leq 1/(8K) + \delta_a$. The innovation lies in an interval of length $1/16$. Within a bin its expected absolute distance from the midpoint is $1/(64M)$. Since changing the scalar innovation by $z - z'$ changes the next state by at most $d|z - z'|$ in the ℓ^1 norm, a certified numerical query has an error allowance

$$e_t \leq \frac{\beta d L_{t+1}}{64M} + \zeta_t. \quad (10)$$

Here ζ_t covers arithmetic, interval evaluation and label rounding. It does not suppress the integration error. The terminal label allowance is e_T .

PROPOSITION 4—Separated-resource policy account: *Under the feasible-witness hypotheses, the returned policy satisfies*

$$0 \leq J^\pi - V^* \leq \frac{a}{N} + \frac{b}{K} + \frac{q}{M} + \eta, \quad (11)$$

where

$$\begin{aligned} a &= d \sum_{t < T} \beta^t L_t + d \beta^T L_g, & b &= \frac{1}{8} \sum_{t < T} \beta^t D_t, \\ q &= \frac{d}{32} \sum_{t < T} \beta^{t+1} L_{t+1}, & \eta &= \sum_{t < T} \beta^t (2\tilde{\xi}_t + D_t \delta_a) + 2\beta^T e_T. \end{aligned} \quad (12)$$

The primitive coefficients are independent of the computed labels. The leading number of innovation evaluations in target formation is

$$T(N+1)^d(K+1)M. \quad (13)$$

The compiled continuation additionally incurs its corner-evaluation and preprocessing costs. Pilot selection, policy extraction, verification, arithmetic bit lengths and output are separate charged components.

PROOF: Insert the state and action radii and (10) into the feasible-witness loss bound. The selected-policy upper account contributes the second copy of e_t , not a second action-search allowance. Collecting the coefficients of N^{-1}, K^{-1}, M^{-1} gives (12). There are $(N+1)^d$ state nodes, $K+1$ actions per node and M innovation representatives per action at each nonterminal date. No independence of fitting errors is used. *Q.E.D.*

The account explains why a common further state rung need not be a diagonal refinement. In the $T=3, p=4, d=2$ cell, the exact primitive coefficients are approximately $a=51.1569, b=2.3650$ and $q=1.0486$. The choice $(N, K, M) = (128, 16, 8)$ has account about $0.6786 + \eta$, below target one when the numerical remainder fits the remaining budget. The diagonal choice $(128, 128, 128)$ is more accurate, but its target-generation innovation count at that same state grid is $129 \cdot 128 / (17 \cdot 8)$ times larger. This is an arithmetic comparison of two declared allocations, not an observation of the unexecuted diagonal service or a runtime ratio. Both witness and conventional FVI receive the separated allocation in the experiment.

4.2. The optimal continuous allocation and its boundary cases

A useful planning problem minimizes the leading product $N^d KM$ subject to (11). This proxy isolates the primitive evaluation term; it is not the entire operation count or the measured wall clock. Write $E = \varepsilon - \eta > 0$ and suppose first that a, b, q are positive and all optimal resources exceed their required lower bounds.

PROPOSITION 5—Error shares and clipped allocation: *The unique interior minimizer of the continuous proxy has*

$$\frac{a}{N} = \frac{d}{d+2}E, \quad \frac{b}{K} = \frac{1}{d+2}E, \quad \frac{q}{M} = \frac{1}{d+2}E. \quad (14)$$

For lower resource bounds $r_i \geq \underline{r}_i > 0$, put $c = (a, b, q)$, $p = (d, 1, 1)$ and $\bar{x}_i = c_i / \underline{r}_i$. If $E < \sum_i \bar{x}_i$, there is a unique positive λ such that

$$x_i = \min\{\bar{x}_i, p_i \lambda\}, \quad \sum_i x_i = E; \quad r_i = c_i / x_i. \quad (15)$$

These resources minimize the constrained continuous proxy. If $E \geq \sum_i \bar{x}_i$, all lower resource bounds suffice. Coordinates with zero error coefficient remain at their resource lower bound. When dyadic lower bounds are used, rounding each optimal resource upward to a power of two preserves the error guarantee and multiplies this proxy by less than 2^{d+2} .

PROOF: Set $x_i = c_i / r_i$. Minimizing the resource product is equivalent to maximizing $\sum_i p_i \log x_i$ over $0 < x_i \leq \bar{x}_i$ and $\sum_i x_i \leq E$. Strict concavity and the multiplier conditions give p_i / x_i equal on unconstrained coordinates and larger on capped coordinates. This is precisely (15). The sum of its right sides is strictly increasing until all caps bind. The unconstrained case yields (14). Upward dyadic rounding never enlarges an error term and multiplies $N^d KM$ by less than the stated factor. Neither argument bounds rational-arithmetic bit cost or computer time. *Q.E.D.*

The optimization is a standard separable resource-allocation calculation. Its role here is to expose the error coefficients of the actual own-future policy construction, including integration and implementation, rather than to claim novelty for the multiplier argument. A finite experimental menu need not contain the continuous optimum. The prototype uses a common, frozen dyadic menu for all generators

and records all of it. It does not describe this menu as a globally optimal tuning policy for either FVI or the witness method.

4.3. A complete tolerance frontier

For method m , let a prospectively fixed ladder return certificates $g_{m,r}$ and cumulative costs $w_{m,r}$, including failed earlier rungs. Define

$$r_m(\varepsilon) = \min\{r : g_{m,r} \leq \varepsilon\}, \quad W_m(\varepsilon) = w_{m,r_m(\varepsilon)}. \quad (16)$$

An empty set is a capped failure, not an interpolated success. Sort the finite union of all certificate values and zero. On each left-closed, right-open interval between successive values, every method's first crossing and complete work are constant. Evaluating the first-crossing rule at the left endpoint therefore reconstructs the entire positive-tolerance frontier exactly, even when successive certificates are not monotone. An unbounded final interval is retained. Repetitions have separate costs, while their mathematical checkpoints must be checked for identity.

This finite partition supplies the appropriate interpretation of a threshold advantage. At a fixed resolution the witness method may cost more. At a tolerance where it crosses one rung earlier, the smaller certificate may more than offset that cost. Whether this happens, and over which tolerance interval, is determined by (16); it is not determined by a favorable single threshold. Under an additional homogeneous model $g_m(n) = A_m/n$ and $w_m(n) = C_m n^\kappa$, the continuous frontier ratio is $(C_W/C_F)(A_W/A_F)^\kappa$. This last expression is a conditional explanation, not a fitted scaling law for the experiments. The release supplies the exact finite partition, sensitivity around one, complete prefixes and separate fixed-resolution observations.

5. THE NONLINEAR INVESTMENT FAMILY

The economic law is the original dimension-normalized investment family. For $x \in [0, 1]^d$, scalar action $0 \leq a \leq \kappa(x)$, and cyclic coordinate $v = j + 1$,

$$\kappa(x) = \frac{1}{8} + \frac{1}{8d} \sum_j x_j, \quad (17)$$

$$F_j(x, a, Z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_v}{8} + \frac{x_j(1 - x_v)}{16} + \gamma_j a + s_j Z,$$

where (γ_j, s_j) alternates between $(1/2, 1)$ and $(1/4, -1)$ and Z is uniform on $[-1/32, 1/32]$. The initial state is uniform on the cube and $\beta = 15/16$. The stage and terminal costs are

$$c(x, a) = \frac{2}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2 \left(\frac{1}{2} - \frac{2}{d} \sum_j x_j \right)_+^2 + a^2 + 4a^4, \quad (18)$$

$$g(x) = \frac{4}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2 \left(\frac{1}{2} - \frac{2}{d} \sum_j x_j \right)_+^2.$$

The original coefficients and continuous laws are unchanged. The domain is invariant: the drift lies in $[1/16, 11/16]$, $a \leq 1/4$, $\gamma_j \leq 1/2$, and the innovation magnitude is at most $1/32$. The results are for this theoretical economy, not calibrated installation fees or estimated welfare magnitudes.

6. TRAINED NEURAL CONTINUATIONS AND EXACT INNOVATION INTEGRATION

The constructive backend supplies a feasible reference without relying on the success of nonconvex optimization. A trained backend addresses a different computational question: whether features fitted to the operator's own continuation targets produce useful candidate actions at a prescribed resource budget. These two backends are not identified with one another. In particular, fitting the hidden weights below is not compilation of the min-plus construction in Theorem 2.

Let the transition have the one-factor form

$$F_t(x, a, Z) = \mu_t(x, a) + s_t(x, a)Z, \quad Z \sim \text{Unif}[-r, r].$$

The deterministic map μ_t may be nonlinear and the state may be multidimensional. Consider a fitted continuation with m hidden units,

$$h(y) = c + v^\top y + \sum_{j=1}^m u_j (w_j^\top y + b_j)_+. \quad (19)$$

Both hidden and output coefficients may be data dependent. Once a candidate has been constructed, its stored coefficients specify an exact numerical function;

uncertainty about whether training found a good function is not an arithmetic uncertainty about the function that was stored.

LEMMA 6—Integrated neural continuation: Put $z_j = w_j^\top \mu_t(x, a) + b_j$ and $R_j = r|w_j^\top s_t(x, a)|$. Then

$$P_t^a h(x) = c + v^\top \mu_t(x, a) + \sum_{j=1}^m u_j \Phi(z_j, R_j), \quad (20)$$

where $\Phi(z, 0) = z_+$ and, for $R > 0$,

$$\Phi(z, R) = \begin{cases} 0, & z \leq -R, \\ (z + R)^2 / (4R), & -R < z < R, \\ z, & z \geq R. \end{cases}$$

For fixed stored coefficients and evaluable primitives, a point expectation uses $O(md)$ arithmetic operations in addition to evaluating μ_t and s_t . Outward evaluation on a state-action box gives a valid enclosure. There is no innovation-grid discretization remainder in this analytic expectation.

PROOF: The scalar random variable $w_j^\top s_t Z$ is uniform on $[-R_j, R_j]$, or is identically zero. Integrating its positive part gives the displayed three cases. Linearity of expectation gives (20); the affine innovation term has mean zero. Each feature requires two inner products and a bounded number of scalar operations. The function Φ is nondecreasing in each argument on $R \geq 0$: in the interior its radius derivative is $(R^2 - z^2)/(4R^2) \geq 0$, and outside the interior it is constant in the radius. Endpoint evaluation with directed arithmetic, enclosing an uncertain branch rather than choosing a favorable branch, therefore preserves containment. Negative output weights must reverse interval endpoints in the usual way. *Q.E.D.*

The formula is a property of this neural class under the stated innovation law, not a general exact-integration result for arbitrary deep networks or unknown kernels. Quadratic critics also have analytic expectations here; tree critics use valid bin enclosures. Thus eliminating quadrature for the neural backend is a specific computational facility, not a claim that conventional approximation necessarily needs

quadrature. On the active hinge region the equivalent expression $(z + R)^2 / (4R)$ avoids subtracting two nearly equal squares. At $R = 0$ the separate branch is essential.

The fitted construction proceeds backward. At each date it draws training states, forms a finite feasible action menu, evaluates current cost plus the integrated *own* subsequent critic, minimizes these labels, and fits the next critic. Its training labels are not evaluations of the optimal Bellman continuation. Fitted value iteration with general function classes has a substantial literature; see [Munos and Szepesvári \(2008\)](#) and the tree-based construction of [Ernst et al. \(2005\)](#). Our use of training is accompanied by a separate whole-cell verification obligation. No training mean-square error is substituted for that obligation.

In the executed neural construction the hidden layer has width 32. All hidden and output coefficients receive 120 full-batch Adam updates, followed by a ridge fit of the affine and output coefficients on the learned features. Coefficients are rounded to 24 binary fractional bits. The verifier evaluates this quantized function, not a hypothetical unrounded optimizer. The fitted hidden weights, targets, states, seeds and quantized output are retained. A fitted nonlinear network, a quadratic continuation, an ExtraTrees continuation, and the two original constructive frontends are therefore identifiable, distinct candidate generators. Whether their additional proposals change the deployed policy is an empirical question that we report separately from their fitting errors.

6.1. *Certified action search for the fitted neural continuation*

The same integrated representation also permits a constructive action search, rather than a second unconverged optimization problem. Fix a reported state in the original economy, so the deterministic next state is $\mu(x, 0) + \gamma a$. Let the feasible deployed actions be $\{0, \Delta, \dots, M\Delta\}$, with $M\Delta$ no greater than the whole-cell lower capacity. Interpret stored coefficients and the reported state as exact dyadic numbers.

THEOREM 7—Algebraic neural action witnesses: *For a continuation of the form (19), its integrated Bellman objective in the original scalar investment action has at most*

2m interior breakpoints. Between consecutive breakpoints it is a quartic polynomial

$$4a^4 + q_2a^2 + q_1a + q_0,$$

with rational coefficients. An exact minimizer on the deployed action lattice can be found among at most $22m + 13$ candidate indices, independently of the number of lattice points. Construction requires $O(md + m^2 + m \log m)$ rational arithmetic and comparison operations, plus $O(m)$ cubic-root isolation problems at the action precision. Coefficient bit lengths and root-isolation work must also be charged. The selected action is an original fitted-critic action witness, not a min-plus encoding.

PROOF: For feature j , write its preactivation mean as $z_j + s_j a$ and its innovation radius as R_j . When $s_j \neq 0$, branch changes occur only at $(\pm R_j - z_j)/s_j$; for $R_j = 0$ these coincide. Retain those breakpoints inside the feasible interval and its two endpoints. Lemma 6 makes each feature quadratic, affine or zero between breakpoints. Adding $a^2 + 4a^4$ and the affine part of the critic gives the displayed polynomial. Its derivative is $16a^3 + 2q_2a + q_1$ and has at most three real roots in each of at most $2m + 1$ pieces.

Between stationary points and breakpoints the objective is monotone. Hence a lattice minimizer lies immediately to either side of a breakpoint or a stationary point, or at a feasible endpoint. Isolate each stationary root in a rational interval of width at most $\Delta/4$ and include every index from the floor of its lower endpoint divided by Δ through the ceiling of its upper endpoint divided by Δ . This contributes at most three indices per isolated root and contains the required neighboring lattice points. The breakpoints and endpoints contribute at most $2(2m + 2)$ indices. Thus $9(2m + 1) + 4m + 4 = 22m + 13$ is a valid bound. Extra indices and duplicate candidates do not invalidate minimization. Exact rational evaluation and lexicographic value-action comparison settle ties.

Computing feature coefficients costs $O(md)$, sorting the breakpoints costs $O(m \log m)$ comparisons, and literal formation of the polynomial on each piece costs $O(m^2)$ operations. Rational isolation of the finitely many cubic roots terminates at positive action precision, including multiple roots. This proves the construction and

the stated separation between arithmetic counts and bit-dependent root-isolation work. Q.E.D.

The solver is implemented with rational root isolation and is checked against exhaustive exact lattice minimization, including saved fitted critics from each of the three state dimensions. It minimizes the stored critic’s objective at the reported state. Whole-cell safety still requires the true-advantage certificate; critic optimality alone does not justify deployment. The preceding R55 economic experiment used its prospectively fixed 17-action construction menu. Its policies and measured clocks are not retroactively replaced by this new action solver, and its cost intervals are not attributed to untested replacement policies.

The new experiment in Section 16 executes the algebraic solver in independently reconstructed prospective services. Its policies, complete work and validation streams are recorded separately from that preceding experiment.

7. A COMPLETE-POLICY CERTIFICATE ON NON-TENSOR ACQUIRED STATES

For the remainder of the new verification and stopping results, the recursion is the bounded finite-horizon expected-cost model. Let $\mathcal{C}$ be a finite measurable partition, with closed boxes used only to verify the cells. Boundary ties follow a specified observation rule. A deployed action is constant on each reported cell and feasible at every state in its closure. A verification box meeting several cells must retain every possible future action. The argument does not require a tensor partition or a continuous actor.

Fix a complete incumbent π^k and define its true advantage

$$A_t^k(x, a) = c_t(x, a) + \beta_t P_t^a J_{t+1}^{\pi^k}(x) - J_t^{\pi^k}(x).$$

The incumbent has advantage zero. For each cell C , let a_i be robustly feasible candidates and let $\mathcal{B}_t(C)$ be a finite interval cover containing every action feasible at every $x \in C$. The cover may include additional actions; its bounds must remain valid on the feasible subset. Suppose verified endpoints satisfy

$$A_t^k(x, a_i) \leq u_i(C), \quad \ell_B(C) \leq A_t^k(x, a) \quad (a \in B \cap A_t(x)).$$

Set $U_t^k(C) = \min\{0, \min_i u_i(C)\}$ and $L_t^k(C) = \min\{0, \min_B \ell_B(C)\}$. Select the candidate attaining U , retaining the incumbent on a tie. All dates are computed against the same old policy and adopted simultaneously.

THEOREM 8—Directed policy improvement: *Under these conditions the new policy is feasible and*

$$J_t^{\pi^{k+1}}(x) \leq J_t^{\pi^k}(x) \quad (21)$$

for every state and date. With $E_t^k = \sup_x (J_t^{\pi^k}(x) - J_t^*(x))$ and $\epsilon_t^k = \max_C (U_t^k(C) - L_t^k(C))$,

$$E_t^{k+1} \leq \beta_t E_{t+1}^k + \epsilon_t^k. \quad (22)$$

For $B_{t,j} = \prod_{i=0}^{j-1} \beta_{t+i}$, $B_{t,0} = 1$, and $m = \min\{K, T - t\}$,

$$E_t^K \leq \sum_{j=0}^{m-1} B_{t,j} \epsilon_{t+j}^{K-1-j} + B_{t,m} E_{t+m}^{K-m}.$$

The final term vanishes when $K \geq T - t$. Exact full-action greedy certificates thus recover the original Bellman optimum within T passes. The coefficient one multiplying the directed error cannot be reduced uniformly.

PROOF: The selected advantage is at most $U \leq 0$. Starting with the common terminal value, backward monotonicity proves (21). The full-action lower cover gives

$$A_t^k(x, \pi_t^{k+1}(x)) - \inf_{a \in A_t(x)} A_t^k(x, a) \leq U_t^k(C) - L_t^k(C).$$

Replacing the new continuation by the old one using safety, and comparing with an arbitrarily near-optimal Bellman action, yields (22). Taking the infimum limit avoids an attainment assumption. Iterating the inequality gives the displayed finite-horizon expansion. For sharpness take one decision date, an incumbent costing $\delta > 0$, and a feasible unproposed action costing zero. An exact incumbent upper endpoint is zero and an exact covering lower endpoint is $-\delta$. Retaining the incumbent gives both error and directed allowance equal to δ . *Q.E.D.*

Theorem 8 restates the original directed result in the partition notation used by the new implementation. It is deliberately representation neutral. Its premise is stronger than checking a finite candidate menu: without the full-action cover, a safe improvement need not have a small gap from the continuous-action optimum. A change of acquisition partition between passes is permitted when the old policy is transferred exactly to its child cells; the complete incumbent remains the same function.

7.1. A one-state residual and value cache

Let $h_T = g$ and h_t be any stored critics, including fitted networks. Write $e_t = J_t^\pi - h_t$ and

$$r_t = c_t(\cdot, \pi_t) + \beta_t P_t^{\pi_t} h_{t+1} - h_t, \quad e_t = r_t + \beta_t P_t^{\pi_t} e_{t+1}.$$

A cellwise residual interval $\mathcal{R}_t(C)$ encloses r_t throughout the cell. If $\mathcal{E}_{t+1}(C')$ encloses the next error, each innovation bin maps a current box and action into a next-state box. The hull of $\mathcal{E}_{t+1}(C')$ over all cells intersecting that box encloses every possible next error. Denote the probability-weighted sum of those hulls by $\mathfrak{P}_t^a \mathcal{E}_{t+1}(C)$. Bin probabilities refer to the original continuous law, not an empirical law of midpoints.

PROPOSITION 9—Cached residual verification: *Starting from $\mathcal{E}_T = \{0\}$, the backward recursion*

$$\mathcal{E}_t(C) = \mathcal{R}_t(C) + \beta_t \mathfrak{P}_t^{\pi_t} \mathcal{E}_{t+1}(C) \tag{23}$$

encloses e_t on every cell. An independently computed direct value interval $\mathcal{V}_t(C)$ may be intersected with $h_t(C) + \mathcal{E}_t(C)$, and $\mathcal{E}_t(C)$ with $\mathcal{V}_t(C) - h_t(C)$. These intersections preserve containment. For feasible a, b the sum

$$c_t(C, a) - c_t(C, b) + \beta_t \{(P_t^a - P_t^b) h_{t+1}(C) + \mathfrak{P}_t^a \mathcal{E}_{t+1}(C) - \mathfrak{P}_t^b \mathcal{E}_{t+1}(C)\}$$

encloses the fixed-incumbent action difference. Intersecting it with the analogous direct-value difference gives a valid directed interval. Identical actions have the exact interval $\{0\}$.

PROOF: For every innovation in a bin, the next state belongs to one of the intersected cells; its error lies in the corresponding stored interval and hence in their hull. Integration against the true bin probabilities preserves order. Backward induction in the residual identity proves (23). Each displayed intersection combines two sets known to contain the same real value. Substituting $J^\pi = h + e$ in the action difference proves the final statement. An empty intersection indicates an invalid numerical calculation and must stop acceptance, rather than licensing a selected midpoint. Q.E.D.

This construction stores one-state bands rather than a table over state pairs. It sacrifices dependence between two trajectories when taking separate hulls; in particular, actor ambiguity and broad residual ranges can make it conservative. For locally Lipschitz critics, a centered residual evaluation can be intersected with the natural interval. A valid generalized-gradient enclosure suffices for the line-segment mean-value bound of a ReLU critic. This is not permission to differentiate a discontinuous acquired actor. On one cell its action is fixed; across cells the cache explicitly retains the alternative actions.

7.2. Signed sensitivity of the installed reference

The original investment family in Section 5 has a smooth zero-investment reference. Its stage and terminal state costs are continuously differentiable, including the squared shortage term. Let μ denote the zero-noise drift and γ the action exposure. For a state box X_0 , propagate boxes X_{j+1} containing $\mu(X_j, 0) + sZ$ for the complete innovation support. Starting from a terminal gradient enclosure $\mathcal{G}_r \supset \nabla g(X_r)$, put

$$\mathcal{G}_j \supset \nabla c(X_j, 0) + \beta D_x \mu(X_j, 0)^\top \mathcal{G}_{j+1}. \quad (24)$$

The inclusion means outward interval multiplication and addition. It is evaluated backward along forward state tubes, with the true sparse Jacobian.

PROPOSITION 10—Signed reference certificate: *The interval $\mathcal{G}_0$ encloses ∇J_0^0 throughout X_0 . For a nonnegative proposed action a , enclose the gradient of the remaining zero policy at every state $\mu(x, \theta a) + sZ$, $0 \leq \theta \leq 1$. If $[v_-, v_+]$ encloses the average*

projection $\mathbb{E}[\gamma^\top \nabla J_{t+1}^0]$ over that segment, then

$$A_t^0(x, a) \in a^2(1 + 4a^2) + \beta a[v_-, v_+]. \quad (25)$$

The signed interval may be intersected with Proposition 9. The construction is valid for the complete zero reference; it does not differentiate a subsequently changed, cellwise constant policy.

PROOF: Differentiate the finite fixed-policy recursion. On the invariant compact domain, the derivatives are bounded, so differentiation and the finite conditional expectations may be interchanged. For every shock path the chain rule has the enclosed Jacobian form in (24); backward induction and expectation preserve the enclosure. The fundamental theorem of calculus along the first action segment gives

$$(P_t^a - P_t^0)J_{t+1}^0(x) = a \int_0^1 \mathbb{E}[\gamma^\top \nabla J_{t+1}^0(\mu(x, \theta a) + sZ)] d\theta.$$

The state-only current costs cancel and the remaining current difference is $a^2 + 4a^4$. This proves (25). Partitioning the first innovation into bins and integrating valid projected gradient hulls with their exact probabilities preserves the same argument. *Q.E.D.*

PROPOSITION 11—Reference stability and computational work: *In the dimension-normalized investment family, for every $d \geq 2$,*

$$\|D_x \mu\|_\infty \leq \kappa := 11/16, \quad \|\nabla c(\cdot, 0)\|_1 \leq 15/2, \quad \|\nabla g\|_1 \leq 10.$$

Consequently, with r stages remaining,

$$\text{Lip}_\infty(J_r^0) \leq \frac{15}{2} \sum_{j=0}^{r-1} (\beta \kappa)^j + 10(\beta \kappa)^r \leq \frac{1920}{91}, \quad \beta = 15/16. \quad (26)$$

For a partition with N leaves, A candidate and covering intervals per leaf, and q innovation bins, the literal cached range-query implementation uses

$$O(TAqN^2d)$$

cell-membership work, in addition to primitive and critic evaluation. Signed reference tubes over a full sweep use $O(ANqdT^2)$ work and can stream their innovation children. Their working state storage is $O(NdT)$ in addition to stored critics, cached bands and retained $O(TAN)$ endpoint records. No q^T recursion or full state-pair table is required.

PROOF: In row j , the only state derivatives of the drift are $1/2 + (1 - x_{j+1})/16$ and $1/8 - x_j/16$, both nonnegative. Their sum is at most $11/16$. The target, cyclic imbalance and shortage terms have gradient ℓ^1 bounds $5/2$, 1 and 4 for current cost; the terminal target bound is 5 . This gives the stated constants. The chain rule yields $L_r \leq 15/2 + \beta\kappa L_{r-1}$ with $L_0 \leq 10$. Its geometric expansion and $\beta\kappa = 165/256$ prove (26).

A queried next-state box is tested against N leaves in d coordinates. There are $O(TANq)$ such root-bin queries. The backward cache therefore has the displayed membership cost. Each gradient tube with r remaining dates visits $O(Nr)$ state rows and $O(Ndr)$ Jacobian coordinates for one first-innovation bin. Summing r across current dates gives $O(ANqdT^2)$. Streaming bins requires storage for one forward tube and its backward gradients, not all future shock combinations. *Q.E.D.*

The stability constant is a bound for this smooth reference, not a dimension-free complexity theorem for solving arbitrary dynamic programs. Resolving a desired policy or all-state accuracy may require the leaf count, network width and action cover to grow. The membership bound explicitly charges a quadratic leaf-count dependence. The executed locally split partition removes a tensor-product representation; it does not prove that a fixed 32-leaf partition resolves every eight-dimensional economic policy.

7.3. What a tighter interval establishes

For fixed candidates, let $C_t(C)$ be the greater of $L_t(C)$ and the minimum of their lower endpoints, including the incumbent's zero. Then

$$U - L = (U - C) + (C - L), \quad U - C \geq 0, \quad C - L \geq 0. \quad (27)$$

The first term measures uncertainty remaining between candidate lower information and the selected upper bound; the second measures the additional continuous-cover allowance. Their maxima need not add to the maximum of their sum. Actor ambiguity, acquisition width, innovation enclosures and arithmetic can enter both terms, so these are not separately estimated causal error components.

More detailed refinement can be recorded without such a causal interpretation. Keep the candidate actions fixed, replace each candidate upper bound by the minimum of its old and refined bounds, and replace the global lower bound by the maximum of its old and refined bounds. Then $U_r \leq U_{r-1}$, $L_r \geq L_{r-1}$, and the certificate improvement at step r is exactly

$$(U_{r-1} - L_{r-1}) - (U_r - L_r) = (U_{r-1} - U_r) + (L_r - L_{r-1}) \geq 0.$$

State subdivision, action-cover subdivision, innovation refinement and higher-precision arithmetic can each supply an independent valid refinement. The present execution records the cache-to-signed-tube refinement and both terms of (27); it does not label an unexecuted six-way refinement experiment as measured evidence.

8. FROM FITTED ACTION WITNESSES TO ECONOMIC POLICY ACCURACY

An exact action witness has a definite but limited meaning: it removes optimization error for the stored continuation at a reported state. The next result identifies the further obligations needed to turn that statement into an error bound for the implemented economic policy. It also isolates a comparison that can be made without attributing the common verifier's contribution to a particular representation.

Fix a complete reference policy π , a date t , and an acquired cell C with representative x_C . Write

$$Q^h(x, a) = c_t(x, a) + \beta_t P_t^a h_{t+1}(x), \quad Q^\pi(x, a) = c_t(x, a) + \beta_t P_t^a J_{t+1}^\pi(x).$$

The superscripts refer to the continuation used, not to different economic primitives. Let G_C be a finite action lattice feasible throughout C , containing the reference action. The reference is assumed implementable on the cell; all selected actions are measurable under the same observation contract. For $g_C \in G_C$, define its reported-

state optimization loss by

$$\kappa_C = Q^h(x_C, g_C) - \min_{b \in G_C} Q^h(x_C, b). \quad (28)$$

This is a nonnegative, finite quantity. Theorem 7 makes it exactly zero for its returned lattice witness, regardless of the training loss or the signs of the learned output weights.

For all $x \in C$ and $a, b \in G_C$, suppose the following nonnegative bounds are available:

$$|[Q^h(x, a) - Q^h(x, b)] - [Q^h(x_C, a) - Q^h(x_C, b)]| \leq \eta_C, \quad (29)$$

$$|[Q^\pi - Q^h](x, a) - [Q^\pi - Q^h](x, b)| \leq \chi_C. \quad (30)$$

Suppose also that every $a \in A_t(x)$ has a $b \in G_C$ with

$$Q^\pi(x, b) \leq Q^\pi(x, a) + \delta_C. \quad (31)$$

These are distinct requirements. The first concerns acquisition within a cell, the second the action-relevant continuation error, and the third the difference between the robust deployed lattice and the original state-dependent continuous feasible set.

Let $[l_C(g_C), u_C(g_C)]$ enclose the true reference advantage $Q^\pi(x, g_C) - J_t^\pi(x)$ throughout C . Let w_C be a verified upper bound on this interval's width. The adoption rule of Theorem 8 is applied to a candidate set containing g_C and the reference, with selected upper endpoint $U_C \leq \min\{0, u_C(g_C)\}$. Denote its complete-action lower endpoint by L_C .

THEOREM 12—Fitted witness transfer: *Under (28)–(31) and the valid whole-cell endpoints above, the selected action a_C satisfies, at every $x \in C$,*

$$Q^\pi(x, a_C) \leq J_t^\pi(x), \quad (32)$$

$$Q^\pi(x, a_C) - \inf_{a \in A_t(x)} Q^\pi(x, a) \leq \min\{U_C - L_C, \rho_C\}, \quad (33)$$

$$\rho_C := \kappa_C + \eta_C + \chi_C + \delta_C + w_C.$$

Consequently the directed finite-sweep recursion remains valid with

$$\varepsilon_t^k = \sup_C \min\{U_t^k(C) - L_t^k(C), \rho_t^k(C)\}.$$

In particular, exact neural action search removes κ_C from this bound, not the other four terms. The coefficient one on the sum ρ_C is sharp without additional restrictions.

PROOF: Fix $x \in C$, an arbitrary feasible action a , and its covering lattice point b . Equations (28)–(30) imply

$$Q^\pi(x, g_C) - Q^\pi(x, b) \leq \kappa_C + \eta_C + \chi_C.$$

Adding (31) gives a bound against $Q^\pi(x, a)$ with the additional term δ_C . The endpoint rule gives

$$Q^\pi(x, a_C) - J_t^\pi(x) \leq U_C \leq u_C(g_C) \leq Q^\pi(x, g_C) - J_t^\pi(x) + w_C.$$

Combining these inequalities and taking the infimum over a proves the ρ_C bound. The full-action lower cover gives the independent $U_C - L_C$ bound. Also $U_C \leq 0$, proving (32). Backward induction from the common terminal value gives safety for simultaneous adoption of all dates. The proof of Theorem 8 then applies with the smaller valid local error; no optimizer attainment for the continuous infimum is needed.

For sharpness, take one decision date, one observation cell containing one state, exact fitted values, and two actions with costs zero and $\delta > 0$. Let the incumbent and only proposed action be the action of cost δ , but let the lattice and full lower cover contain both actions. Then $\kappa_C = \delta$, the other four terms are zero, and the retained policy has loss exactly δ . Thus the coefficient multiplying the unrestricted sum cannot be smaller than one. Q.E.D.

The width term records the conservatism of certifying the proposed action and then choosing among upper endpoints. It is necessary: the action with the smallest upper bound need not have the smallest true advantage. A separately verified bound on $u_C(g_C) - [Q^\pi(x, g_C) - J_t^\pi(x)]$ can replace the full interval width. Like-

wise, sharper contrast bounds may replace (29) and (30). Their validity must be established before they are used in the policy account.

8.1. A primitive continuous-action allowance

The continuous-action condition is explicit in the investment family. Let $\bar{a}(x)$ be its capacity, Δ the action spacing, and

$$m_C \Delta = \Delta \left\lfloor \frac{\inf_{x \in C} \bar{a}(x)}{\Delta} \right\rfloor, \quad G_C = \{0, \Delta, \dots, m_C \Delta\}.$$

If $Q^\pi(x, \cdot)$ is uniformly L_a -Lipschitz, then

$$\delta_C = L_a \{\text{osc}_C \bar{a} + \Delta\} \quad (34)$$

is admissible. For the smooth installed zero policy in the original normalized family, one may take

$$L_a = \frac{3}{4} + \frac{900}{91}. \quad (35)$$

PROOF: For a feasible a , use the smaller of its downward lattice rounding and $m_C \Delta$. Its distance from a is at most $\text{osc}_C \bar{a} + \Delta$, proving (34). The derivative of $a^2 + 4a^4$ has absolute value at most $3/4$ on $[0, 1/4]$. Proposition 11 bounds the zero-policy value's Lipschitz constant in the maximum norm by $1920/91$. The action exposure has maximum norm $1/2$ and the discount is $15/16$. Therefore the continuation's action Lipschitz constant is at most $900/91$, proving (35). *Q.E.D.*

The primitive allowance is valid but can be conservative. The executed policy bound uses the direct continuous-action lower cover, not a claim that the primitive sum is always tighter. The transfer theorem permits both accounts and shows exactly where an improved action search can enter the original Bellman-accuracy objective.

8.2. Nested proposals and identifiable verification gains

Consider two candidate sets $M_0 \subseteq M_1$ for the same reference, acquired cells, complete-action lower cover and valid upper-endpoint procedure. The endpoints of

shared actions must be identical; this condition prevents changes in a fitted residual cache from being confused with changes in the proposed actions. Both sets retain the reference with exact upper advantage zero.

COROLLARY 13—Nested-candidate certificate: *Let $U_i(C)$ be the minimum candidate upper endpoint for M_i . Then*

$$U_1(C) \leq U_0(C) \leq 0, \quad 0 \leq U_1(C) - L(C) \leq U_0(C) - L(C).$$

The improvement is strict precisely when an added candidate has upper endpoint strictly below $U_0(C)$. Both returned complete policies are statewise safe relative to the reference. These inequalities alone do not order their actual expected costs.

PROOF: Taking a minimum over a larger finite set proves the first inequalities. The valid full-action lower bound is no greater than every feasible action's advantage and hence no greater than either selected upper bound. This proves the nonnegative gap inequalities and the strictness criterion. Safety follows from Theorem 8. To see the last distinction, consider a single state and date, an incumbent of cost 2, an old candidate of cost 1 with advantage enclosure $[-1, -1/2]$, and a new candidate of cost $5/4$ with exact advantage $-3/4$. The new candidate has the smaller upper endpoint and is adopted, although its cost is higher than that of the old candidate. Both improve the incumbent. A lower cover of -1 makes both certificates valid. Q.E.D.

In the new experiment the ReLU-exact mode retains every ReLU-menu action and adds the algebraic witness. Both use the same fitted coefficients at a given construction attempt and a critic-independent smooth-reference verifier. Thus the corollary is an observed, testable comparison rather than a representation claim. Exact center-objective regret, stricter upper endpoints, changed deployed actions and independent expected-cost differences are reported separately.

8.3. A paid decision between returned policies

Let $[l, u]$ cover $J(\pi_1) - J(\pi_0)$ on a simultaneous event after both policies have been selected. If adopting π_1 rather than π_0 incurs an additional replacement

charge τ and measured computational work difference ΔW , its net cost difference at computation price $\lambda \geq 0$ lies in

$$[l + \tau + \lambda \Delta W, u + \tau + \lambda \Delta W]. \quad (36)$$

A negative upper endpoint supports adoption; a positive lower endpoint rules it out on the same event. Between those regions the comparison is unresolved. This affine transformation is valid for every declared nonnegative price on that event. It does not turn a normalized benchmark into an estimated installation fee. It links a particular fitted-action computation to the same economic policy and resource question as the original NBO construction.

8.4. Explicit moduli for the trained neural representation

The transfer bound need not leave acquisition error as an unspecified constant. In the original investment family, write the learned continuation as

$$h(y) = v^\top y + c + \sum_{j=1}^m u_j \sigma(w_j^\top y + b_j).$$

Let $s_j = w_j^\top \gamma$, let R_j be the feature's scalar-innovation radius from Lemma 6, and set $D_C = \text{diam } G_C$ and $r_C = \sup_{x \in C} \|x - x_C\|_\infty$. Define

$$d_j(D) = \begin{cases} 0, & s_j = 0, \\ \min\{1, |s_j|D/(2R_j)\}, & s_j \neq 0, R_j > 0, \\ 1, & s_j \neq 0, R_j = 0. \end{cases}$$

All these quantities are determined by the stored coefficients and the declared primitive law.

PROPOSITION 14—Action-sensitive neural moduli: *For the original separable current action cost and drift with maximum-norm Lipschitz constant $11/16$, the acquisition allowance*

$$\eta_C = \beta_t \frac{11}{16} r_C \sum_{j=1}^m |u_j| \|w_j\|_1 d_j(D_C) \quad (37)$$

satisfies (29). In particular, the affine part and every exactly action-null ridge contribute zero to this allowance. For the installed zero reference, a valid continuation-error allowance is

$$\chi_C = \beta_t D_C \left\{ \frac{960}{91} + |v^\top \gamma| + \sum_{j=1}^m |u_j s_j| \right\}. \quad (38)$$

Either allowance may be replaced by the minimum of itself and any independently valid sharper enclosure. Computing the displayed bounds requires $O(md)$ arithmetic, with outward rounding when evaluated numerically.

PROOF: Write the integrated ridge as $\Phi_R(z) = \mathbb{E}\sigma(z + RZ)$ for Z uniform on $[-1, 1]$. For $R > 0$, its derivative is zero, then affine with slope $1/(2R)$, then one. Hence the function

$$z \mapsto \Phi_R(z + s_j a) - \Phi_R(z + s_j b)$$

is Lipschitz with constant at most $\min\{1, |s_j(a - b)|/(2R)\}$. For $R = 0$ the difference of the two hinge derivatives is bounded in absolute value by one almost everywhere, giving the same conclusion with the declared convention. For $s_j = 0$ the function is identically zero. The preactivation mean varies by at most $\|w_j\|_1(11/16)\|x - x_C\|_\infty$. The affine continuation contrast is $v^\top \gamma(a - b)$, which does not depend on x . The state-only current cost cancels, and the current action-cost difference also does not depend on x . Summing the ridge contributions and discounting proves (37).

Under a common innovation, the zero-policy continuation contrast is bounded by $(1920/91)\|\gamma\|_\infty|a - b| = (960/91)|a - b|$. The one-Lipschitz hinge gives a fitted continuation contrast bound $\{|v^\top \gamma| + \sum_j |u_j s_j|\}|a - b|$. The triangle inequality and discount give (38). Independently valid upper bounds may be minimized because each bounds the same nonnegative quantity. The arithmetic count follows from computing the feature projections and norms. Q.E.D.

These formulas connect feature geometry to acquired-state accuracy without assuming that training discovers a null direction. They complement the confidence-qualified learned-direction result: estimated primitive coefficients must still range over the corresponding confidence set. In the present known-law action-search

experiment, the direct signed reference enclosure determines adoption and the full-action lower cover determines the displayed Bellman error. The primitive moduli supply an additional certified account, not a retroactive claim that their conservative sum was the stopping criterion.

9. LOSSLESS SEARCH FOR TRAINED NEURAL ACTION WITNESSES

The fitted-witness transfer result separates an optimization error from the errors in acquisition, continuation evaluation and full-action verification. An exact solver eliminates the first error, but exactness alone does not make the solver economical. We now exploit the learned features to reduce the work of returning precisely the same action witness. The continuation, economic primitives and original Bellman-accuracy target remain unchanged.

9.1. *Exact reduction of integrated neural features*

Fix a state and a stored affine-ReLU continuation. Interpret its finite binary coefficients as rational numbers. With a scalar uniform innovation, the expectation of a ridge is

$$\phi_r(z) = \begin{cases} ((z+r)_+^2 - (z-r)_+^2)/(4r), & r > 0, \\ z_+, & r = 0. \end{cases}$$

The radius r is the absolute projection of the innovation support on the ridge direction. After removing a constant independent of the action, the fitted objective in the investment model takes the form

$$F(a) = q_1 a + q_2 a^2 + 4a^4 + \sum_{j=1}^m \omega_j \phi_{r_j}(z_j + s_j a). \quad (39)$$

All coefficients are rational. Negative fitted weights are permitted. No convexity of the fitted continuation is assumed. The robust feasible action lattice is $\mathcal{L} = \{k\delta : 0 \leq k \leq K\}$, where δ and $K\delta$ are fixed by the acquisition contract.

A feature can be simplified before any lattice search. Let $[v_j^-, v_j^+]$ be the exact range of $z_j + s_j a$ on $[0, K\delta]$. If $v_j^+ \leq -r_j$, its contribution vanishes. If $v_j^- \geq r_j$, it is affine. If $r_j > 0$ and $[v_j^-, v_j^+] \subseteq [-r_j, r_j]$, it is quadratic. Absorb these contributions

into a rational constant c_0 and the coefficients of a and a^2 . Only features crossing an activation-region boundary remain in the ridge sum. Consequently

$$F(a) = c_0 + \tilde{q}_1 a + \tilde{q}_2 a^2 + 4a^4 + \sum_{j \in \mathcal{C}} \omega_j \phi_{r_j}(z_j + s_j a) \quad (0 \leq a \leq K\delta). \quad (40)$$

This is an identity, not pruning by a small fitted weight. A feature with a large weight may be eliminated exactly, whereas a small crossing feature remains unless its whole contribution is enclosed.

9.2. An interval screen with an exact return

Write $f_k = F(k\delta) - c_0$. Compute outward enclosures $[\ell_k, u_k]$ of all f_k and put

$$b = \min_{0 \leq k \leq K} u_k, \quad S = \{k : \ell_k \leq b\}. \quad (41)$$

Compare the original objective exactly on S , choosing the smallest index on a tie. The weak inequality in the definition of S is essential. Floating-point endpoints reject a point only when they prove that it cannot minimize the original objective; they never supply an approximate answer in place of the exact comparison.

THEOREM 15—Lossless neural action search: *Suppose the intervals in (41) are finite, ordered and valid. Then S contains every exact lattice minimizer. Exact rational comparison on S , with the same smallest-index convention, returns exactly the witness of exhaustive lattice minimization and of the terminating algebraic solver.*

Let $\eta = \max_k (u_k - \ell_k)$. Every survivor satisfies

$$F(k\delta) - \min_{a \in \mathcal{L}} F(a) \leq 2\eta. \quad (42)$$

In particular, a unique minimizer is the only survivor whenever its gap to the second-best lattice point exceeds 2η .

More generally, let $\mathcal{M} \subseteq \mathcal{L}$ be the exact minimizer set. If, for some $\mu > 0$,

$$F(a) - \min_{\mathcal{L}} F \geq \frac{\mu}{2} \text{dist}(a, \mathcal{M})^2 \quad (a \in \mathcal{L}),$$

then

$$|S| \leq \min \left\{ K + 1, |\mathcal{M}| \left(1 + 2 \left\lfloor \frac{2\sqrt{\eta/\mu}}{\delta} \right\rfloor \right) \right\}.$$

The growth condition is an additional assumption, not a property inferred from a fitted ReLU representation.

PROOF: Let k_* minimize f_k . For every j , $f_{k_*} \leq f_j \leq u_j$, hence $\ell_{k_*} \leq f_{k_*} \leq b$ and $k_* \in S$. This argument retains all minimizers, including ties. Exact comparison with the same tie rule therefore returns the same index. The algebraic solver enumerates lattice neighbors of every activation boundary and isolated stationary point on each polynomial piece; its established exhaustive-search equivalence then gives the same witness.

For a survivor k , validity and the width bound imply

$$f_k \leq \ell_k + \eta \leq b + \eta \leq u_{k_*} + \eta \leq f_{k_*} + 2\eta.$$

This proves (42) and the unique-gap statement. Under the additional growth condition a survivor is within $2\sqrt{\eta/\mu}$ of some element of $\mathcal{M}$. There are at most $1 + 2\lfloor 2\sqrt{\eta/\mu}/\delta \rfloor$ lattice points within that distance of one lattice minimizer. Taking a union, and then the total lattice cardinality, proves the last bound. *Q.E.D.*

The generic screening implication uses only interval order. The implementation contribution here is the exact reduction of integrated trained ridges, outward evaluation of their reduced objective, and exact recovery of the original NBO witness. On a failed arithmetic check or a lattice exceeding the declared screening limit, the implementation calls the original terminating algebraic solver. A fallback is charged as work and cannot be recorded as an approximate successful screen. A wide but valid enclosure may retain the entire lattice; the theorem promises correctness in that case, not a speed gain.

PROPOSITION 16—Arithmetic work and working storage: Let $N = K + 1$, let $m_c = |C|$, and let $s = |S|$. Forming the state-specific neural coefficients and performing exact region reduction requires $O(dm)$ arithmetic operations for an m -ridge continuation in

dimension d . The screen uses $O(N(1 + m_c))$ fixed-precision interval operations, followed by $O(sm)$ rational operations to compare survivors using the original objective. The vectorized implementation uses $O(N(1 + m_c))$ interval working storage, in addition to the fitted coefficients, and returns $O(s)$ retained indices. These are arithmetic counts; the cost of rational arithmetic also depends on operand bit lengths.

PROOF: Each ridge requires state and innovation projections and a constant number of exact range tests. Each remaining ridge contributes one scalar interval expression at each lattice point. Pairwise outward summation of the ridge contributions is linear in their number. The exact return evaluates the original m -ridge objective at each survivor. The implementation retains ridge-by-lattice batches, giving the stated storage rather than a streaming bound. Integer index comparison resolves equal rational values. None of these counts eliminates the dependence on d , lattice resolution or fitted width. Q.E.D.

Terminal optimization is unchanged in the experiment. The screened intervention concerns only nonterminal trained-ReLU continuations. In particular, time saved by an analytic terminal formula is not credited to learned-feature reduction.

9.3. Preservation of a prospective economic return

A solver is used inside a procedure, not in isolation. Consider a deterministic prospective controller conditional on its construction data and random streams. At each attempted resolution it constructs an acquired policy, verifies whole-cell advantages and continuous-action lower covers, and then runs the declared sequence of cost-inference looks. It returns at the first certified economic target, advances resolution when required by its rule, or reports budget exhaustion. Suppose its branching decisions depend on these mathematical outputs, not on elapsed time or solver-specific diagnostic counters.

THEOREM 17—Prospective transcript invariance: *Run the same controller with the algebraic solver and the lossless screened solver. Suppose corresponding construction steps produce identical stored continuations, acquisition partitions and other candidate actions, and use identical inference streams. Then both controllers return the same policy,*

1 *have identical whole-cell verification endpoints and cost-inference endpoints, and stop at* 1
 2 *the same resolution and inference look with the same status. Their exact fitted-optimization* 2
 3 *loss is the same, namely zero on the declared action lattice. Their all-state Bellman-error* 3
 4 *certificate is unchanged.* 4

5 *The returned policies have exactly equal expected economic cost under every admissible* 5
 6 *initial distribution. This equality does not require statistical inference. It does not assert* 6
 7 *equality of computation cost.* 7

8 8
 9 PROOF: Condition on the common construction inputs and streams. Theorem 15, 9
 10 or the exact fallback, gives identical proposed action indices at every leaf and date. 10
 11 The remaining candidate menu and its ordering are the same. The same verifier 11
 12 therefore evaluates the same intervals and resolves the same incumbent-preserving 12
 13 comparisons, producing an identical acquired policy and continuous-action error 13
 14 account. Evaluating that policy on the same interval paths gives identical path 14
 15 endpoints, hence identical inference bounds and the same next controller deci- 15
 16 sion. Induction over inference looks and attempted resolutions proves the entire 16
 17 transcript assertion. No additional attempt occurs after a successful return. The 17
 18 inherited fitted-witness transfer and directed policy-improvement theorems apply 18
 19 to the identical witnesses and certificates. Finally, identical policies have the same 19
 20 controlled law and expected cost for every initial law. Q.E.D. 20

21 21
 22 The theorem is conditional on actual parameter identities, not merely on a shared 22
 23 training seed. The empirical audit checks those identities. It does not apply to a 23
 24 wall-clock deadline rule: a faster solver can reach a different stage before a real-time 24
 25 deadline. Nor does a target that is not attained under the declared finite budgets 25
 26 become impossible in the underlying economic model. 26

27 27
 28 COROLLARY 18—Computation prices at an unchanged policy: *Suppose both* 28
 29 *implementations return π and face specified computation charges C_A, C_S . At a nonnegative* 29
 30 *price λ , their total expected policy-plus-computation cost difference is* 30

$$\{J(\pi) + \lambda C_S\} - \{J(\pi) + \lambda C_A\} = \lambda(C_S - C_A).$$

If an installed controller has a verified replacement-gain interval $[l, u]$, a fixed installation charge τ gives net gain in $[l - \tau - \lambda C_i, u - \tau - \lambda C_i]$ for implementation i .

PROOF: Cancel the identical policy costs in the first display and subtract the specified deterministic charges from the gain interval in the second. *Q.E.D.*

This comparative statement holds for specified resource charges. Three measured process repetitions do not estimate a distribution-free expected runtime, and the study does not calibrate λ or τ . They do, however, permit a directly interpretable work comparison at exactly the same economic and statistical accuracy, without attributing different welfare to identical neural policies.

10. EXACT ACTION RECOVERY AND THE BELLMAN ACCURACY CONTRACT

The preceding lossless screen makes the number of exact comparisons depend on the separation of the best lattice actions. That dependence is useful when separation is substantial, but it is not a reason to assume a favorable regime. We now exploit the integrated objective directly. The resulting exact solver does not require a growth constant, a unique minimizer, or isolation of an algebraic root. We also separate this local optimization question from verification against the original Bellman optimum.

10.1. Finite differences instead of root isolation

Fix a state and a stored continuation. After the exact activation reductions, a scalar-action objective under a finite mixture of scalar uniform innovations has the form

$$F(a) = c_1 a + c_2 a^2 + c_4 a^4 + \sum_{j=1}^m w_j \Phi_{r_j}(z_j + s_j a) + \sum_{j=1}^h v_j (y_j + t_j a)_+^2, \quad a \in [0, \bar{a}], \quad (43)$$

up to a common constant. Here $c_4 \geq 0$, all stored coefficients are rational, and $\Phi_r(z) = \mathbb{E}(z + U)_+$ for U uniform on $[-r, r]$, with $\Phi_0(z) = z_+$. The original investment model has $c_4 = 4$. Mixture probabilities and shifts can be absorbed into the ridge weights and intercepts. Negative weights are allowed. Breakpoints are rational and divide the interval into at most $K \leq 2m + h + 1$ polynomial pieces. On each

closed piece, continuity gives the polynomial $P(a) = d_0 + d_1a + d_2a^2 + c_4a^4$ at the endpoints as well.

Let $p_j = P(j/Q)$ on the consecutive feasible indices $j = l, \dots, u$ of one piece. Define $d_j = p_{j+1} - p_j$. Direct expansion gives

$$d_j = \frac{d_1}{Q} + \frac{d_2(2j+1)}{Q^2} + \frac{c_4(4j^3 + 6j^2 + 4j + 1)}{Q^4}, \quad (44)$$

$$d_{j+1} - d_j = \frac{2d_2}{Q^2} + \frac{c_4(12j^2 + 24j + 14)}{Q^4}. \quad (45)$$

The last expression is nondecreasing for $j \geq 0$, including when $d_2 < 0$. This is a discrete curvature property, not global convexity of the fitted objective.

LEMMA 19—Exact minimization of one piece: *Suppose $0 \leq l \leq u$ and $c_4 \geq 0$. The smallest minimizing index of $p_l, \dots, p_u$ can be recovered using $O(1 + \log(u - l + 1))$ exact rational sign queries and at most four exact objective comparisons. If every second difference is negative, endpoints suffice. Otherwise let j_0 be the first index with nonnegative second difference. In addition to the endpoints, it suffices to retain j_0 and the first $k \geq j_0$ with $d_k \geq 0$, when that index exists. If it does not exist, the right endpoint is the minimum of the remaining tail.*

PROOF: A sequence with decreasing first differences is discretely concave and has a minimum at an endpoint. If j_0 exists, the prefix through p_{j_0+1} is concave, whereas the tail beginning at p_{j_0} is convex. A minimum of the full sequence is therefore either the left endpoint or a minimum of that convex tail. The latter is at the first nonnegative first difference, or at the right endpoint if all its first differences are negative. The first such index is the smallest tail minimizer, including a run of exact ties. Monotonicity of (45) locates j_0 by integer bisection, and monotonicity of d_j on the tail locates k in the same way. Intervals with fewer than three indices are handled by their endpoints. Exact comparisons of the retained values complete the argument. Q.E.D.

THEOREM 20—Root-free exact action witness: *For (43) on $\{j/Q : 0 \leq j \leq \bar{j}\}$, with $\bar{a} = \bar{j}/Q$, form the rational activation partition, intersect each piece with the integer lattice,*

and apply Lemma 19. Evaluate the original objective at the union of the retained indices, choosing the smallest index on an exact tie. The result is precisely the original lattice minimizer, without a separation assumption or an algebraic root calculation.

A direct implementation uses

$$O((m+h)^2 + (m+h)\log(m+h+1) + K\log(\bar{j}+2)) \quad (46)$$

rational operations and comparisons, and $O(m+h+K)$ stored rational numbers and indices. These are operation counts, not unit-cost bit bounds. If b bounds the bit lengths of all operands and $C_{\text{rat}}(b)$ bounds one rational operation or comparison, multiplying (46) by $C_{\text{rat}}(b)$ is a valid bit-cost account. For input coefficient bit length at most B , fixed degree and denominator Q , a conservative bound is

$$b = O((m+h+1)(B + \log(Q+1) + \log(\bar{j}+2) + \log(m+h+2))).$$

PROOF: The activation formulas are exact on every piece. Its lattice intersection is the consecutive interval from the ceiling of its left endpoint times Q to the floor of its right endpoint times Q ; empty intersections are skipped. These intersections cover every lattice action. The lemma retains a minimizing index for every nonempty intersection and its smallest minimizer. Exact comparison under the original objective is unaffected by the piecewise constants or the earlier common-constant reduction, and gives the original tie rule globally.

Sorting breakpoints costs $O((m+h)\log(m+h+1))$ comparisons. Constructing the polynomial coefficients independently in each piece costs $O(K(m+h))$ operations. The integer bisections cost $O(K\log(\bar{j}+2))$. There are $O(K)$ final candidate indices, and direct evaluation of the original objective costs $O(K(m+h))$. The implementation stores the features, breakpoints and candidate indices, rather than an array indexed by every lattice point. Finally, products of a fixed number of input rationals have bit length proportional to the input lengths. Summing at most $O(m+h)$ such terms admits a common denominator formed by their product; this gives the stated conservative operand bound, including powers of Q , the action-index bound, and the rational breakpoint calculations. Q.E.D.

The theorem is useful in precisely the case where interval separation is inconclusive. A flat piece, a tie between consecutive actions, or a large signed cancellation does not prevent exact bisection. It does not assert that bisection is always fastest: when the lattice is short relative to the activation partition, native exhaustive evaluation can be cheaper. The ablation below therefore includes both, as well as the original symbolic solver after the same exact reduction. Discrete curvature and verified interval exclusion are established numerical principles (Murota, 2003, Moore et al., 2009). The result here is their explicit, root-free realization for the integrated NBO objective, with an exact original-witness contract and a bit-aware implementation account.

10.2. Several controls and uniform-box shocks

The lossless witness contract itself need not be scalar. Let $a \in \mathbb{R}^p$ belong to a bounded rational polytope and to a declared rational action lattice. For a ridge of a one-hidden-layer ReLU continuation, affine exposure gives an argument $z + s^\top a + \sum_{i=1}^q U_i$, where the U_i are independent and uniform on $[-r_i, r_i]$. Zero radii are removed. Finite mixtures of such boxes are allowed, with nonnegative rational probabilities summing to one.

PROPOSITION 21—Exact integrated ridge: For $q \geq 1$ and $r_i > 0$,

$$\mathbb{E}\left(z + s^\top a + \sum_{i=1}^q U_i\right)_+ = \frac{\sum_{\epsilon \in \{-1,1\}^q} \left(\prod_{i=1}^q \epsilon_i\right) \left(z + s^\top a + \sum_{i=1}^q \epsilon_i r_i\right)_+^{q+1}}{2^q (q+1)! \prod_{i=1}^q r_i}. \quad (47)$$

Consequently each fitted objective is rational at every feasible lattice action. Its value and valid box enclosures can be computed with signed output weights and exact ties. Evaluation costs include the 2^q terms per ridge; the formula does not remove shock-dimension dependence.

PROOF: An iterated antiderivative of x_+ of order q is $x_+^{q+1}/(q+1)!$. Apply the fundamental theorem of calculus in each integration coordinate and divide by

the uniform-box volume. Each boundary choice supplies the corresponding sign in (47). Rationality follows from the rational primitives and lattice. The expected hinge is monotone in its scalar argument, so evaluating it at the exact minimum and maximum of $z + s^\top a$ on a rectangle supplies a valid enclosure. A negative output weight reverses the endpoints. Addition and nonnegative mixture weighting preserve inclusion. Q.E.D.

The executed two-control extension uses shared capacity $a_1 + a_2 \leq 1/8$, exposure columns $(1/2, 1/4)$ and $(1/4, 1/2)$, and the original first shock together with an optional independent second shock. The extension is stated rather than conflated with the scalar experiment: setting $a_2 = 0$ and disabling the second shock recovers the original action-dependent objective. The quartic lemma is used only in its stated scalar, piecewise-quartic subclass. The two-shock and two-control calculation instead uses verified box subdivision and exact feasible-point comparison.

THEOREM 22—Finite constrained recovery with a work budget: *Suppose a finite rational action lattice is intersected with a bounded rational polytope, and every visited integer rectangle has a valid lower objective bound. Maintain an exactly evaluated feasible incumbent and a fixed lexicographic tie rule. Discard a rectangle only if it is infeasible, its lower bound is strictly greater than the incumbent value, or equality holds and its smallest possible index is not earlier than the incumbent. Split any other nonsingleton rectangle into two disjoint integer rectangles, and evaluate singleton actions exactly. Then exhaustive subdivision returns the original smallest minimizing feasible action. Switching after any declared node budget to an exact exhaustive solver preserves this conclusion and charges the preceding work in addition to the fallback.*

For side cardinalities $N_1, \dots, N_p$, a depth-first implementation has depth at most $\sum_i \lceil \log_2 N_i \rceil$ under midpoint splitting. Its rectangle stack stores $O(p \sum_i \log(N_i + 1))$ integers. Without pruning it can still visit $O(\prod_i N_i)$ nodes. Objective arithmetic, rational bit lengths and exact-polytope feasibility work must be added to these combinatorial counts.

PROOF: Each split partitions the remaining integer indices without omission. A discarded infeasible rectangle contains no admissible competitor. Strict lower-bound exclusion removes no minimizer that could improve the incumbent. In the

equality case no point can improve its value, and the index condition excludes an earlier tied minimizer. Every unpruned feasible singleton is compared exactly. Finite subdivision therefore retains the globally best value and the specified first minimizer. An exact terminating fallback solves the same remaining mathematical problem, whether or not it reexamines previously visited points. Its repeated work affects cost, not validity. Midpoint splitting halves the active side range, yielding the depth and stack bounds; a complete binary partition has fewer than twice as many nodes as its full rectangular lattice has points. *Q.E.D.*

Replacing a solver by either theorem preserves the earlier prospective transcript whenever that controller depends on the mathematical witness rather than elapsed time or solver diagnostics. This transfers exactness to the acquired policy and its certificates; it does not imply that a fitted-optimal action is optimal for the true continuation. That latter error remains in the existing transfer theorem and is examined by an independent all-state calculation next.

10.3. *A direct certificate against the original Bellman optimum*

Let $\mathcal{C}_t$ be a finite cover of the original state domain with an unambiguous ownership rule for shared boundaries. Starting from terminal bounds on g , maintain a piecewise lower function L_{t+1} for the Bellman value and a piecewise upper function U_{t+1} for the value of an explicitly constructed continuation policy. On a cell C , let $\mathcal{B}_t(C)$ cover every action feasible at every state in C , and let $\mathcal{A}_t(C)$ contain only actions feasible throughout C . Suppose verified endpoints satisfy

$$\ell_t(C, B) \leq c_t(x, a) + \beta_t P_t^a L_{t+1}(x) \quad (x \in C, a \in B \cap \mathcal{A}_t(x)),$$

$$u_t(C, b) \geq c_t(x, b) + \beta_t P_t^b U_{t+1}(x) \quad (x \in C, b \in \mathcal{A}_t(C)).$$

A lower cover may include additional infeasible actions, but every evaluated extension must retain a valid lower enclosure on its feasible subset. All reached state cells and the full original innovation law are included in the expectations. Define

$$L_t(C) = \min_{B \in \mathcal{B}_t(C)} \ell_t(C, B), \quad U_t(C) = \min_{b \in \mathcal{A}_t(C)} u_t(C, b),$$

and implement the minimizing upper candidate on C .

THEOREM 23—Whole-domain Bellman bracket: *The constructed complete policy π is feasible and satisfies*

$$L_t(C) \leq J_t^*(x) \leq J_t^\pi(x) \leq U_t(C), \quad x \in C. \quad (48)$$

Hence $\max_C \{U_t(C) - L_t(C)\}$ is an all-state upper bound on loss against the original continuous-action Bellman optimum. A declared tolerance is attained when its outward bound passes the corresponding comparison; no initial distribution or reference-zero gain is needed.

For a refining family, suppose the exact lower and upper endpoint maps at each fixed approximation level are order preserving and β_t -nonexpansive in the supremum norm. Suppose their one-step errors at the true next Bellman value, together with uniformly bounded numerical endpoint errors on the relevant input range, are at most η_t , inclusive of robust-action approximation and state, action and innovation covering. If the terminal lower and upper errors are each at most η_T , then

$$\sup_C \{U_t(C) - L_t(C)\} \leq 2 \sum_{j=t}^T \left(\prod_{i=t}^{j-1} \beta_i \right) \eta_j.$$

In particular, uniformly vanishing such allowances give arbitrarily small Bellman brackets in finite horizon. The runtime certificate is (48); the consistency statement does not require the algorithm to observe J^* .

PROOF: Assume the next-date inequalities. Monotonicity of each action kernel gives $P_t^a L_{t+1} \leq P_t^a J_{t+1}^*$. The complete action cover and the lower enclosure therefore imply $L_t(C) \leq \inf_{a \in A_t(x)} \{c_t(x, a) + \beta_t P_t^a J_{t+1}^*(x)\} = J_t^*(x)$. The chosen upper action is feasible at every state in C . Its continuation policy has value at most U_{t+1} , so its current value is at most the chosen upper endpoint $U_t(C)$. Feasibility gives $J_t^* \leq J_t^\pi$. Backward induction proves the bracket.

For consistency, let e_t^- bound $J_t^* - L_t$ and let e_t^+ bound $U_t - J_t^*$. Nonexpansivity of the fixed-cover endpoint maps bounds the propagated next-date discrepancy by β_t times its supremum norm. Adding the uniformly budgeted local and numerical

endpoint errors consequently gives $e_t^- \leq \eta_t + \beta_t e_{t+1}^-$ and $e_t^+ \leq \eta_t + \beta_t e_{t+1}^+$, with terminal errors at most η_T . Adding the two unrolled inequalities proves the bound. Primitive uniform continuity and consistent enclosure rules supply vanishing allowances on compact domains when the robust feasible menus and full-action covers converge. These are hypotheses for consistency, not an uncharged value-function oracle. *Q.E.D.*

The numerical implementation uses exact rectangle extrema of stored outward bounds. At the last date it integrates the original terminal cost analytically. At earlier dates each innovation bin retains its exact original probability, and no midpoint is substituted for the bin's conditional expectation. This calculation is a conventional constructive benchmark in the same NBO economy. It puts an attained all-state accuracy and the work paid for it beside the learned prospective services, rather than presenting improvement relative to zero investment as a solution of the Bellman problem.

10.4. *The economic value of an exact implementation change*

For two implementations returning the identical policy, let w_0, w_1 be their measured per-service work, k_0, k_1 their one-time compilation or installation work, and $\lambda \geq 0$ the price of the recorded work unit. Over H uses, with an additional implementation charge I , the new-minus-old total cost is

$$\lambda \{H(w_1 - w_0) + (k_1 - k_0)\} + I.$$

When $w_1 < w_0$, the recorded computation saving pays for a positive setup difference only beyond the corresponding break-even use count. This condition is reported separately from policy-cost comparisons and from the cold single-use boundary. No electricity tariff, processor frequency, installation price or external welfare calibration is inferred from a hosted-runner clock.

10.5. *Common-reference cancellation at the original optimum*

The width of an absolute value box can be dominated by state variation common to all actions. Subtracting the same reference function from both Bellman endpoints

removes that variation without changing the decision problem. This is a use of the original policy-relevant error principle, not a replacement of the Bellman objective by a zero-policy comparison.

Fix any feasible complete reference π^0 and write $h_t = J_t^{\pi^0}$. Define its true reference advantage by

$$A_t^0(x, a) = c_t(x, a) + \beta_t P_t^a h_{t+1}(x) - h_t(x).$$

Set $e_t^*(x) = J_t^*(x) - h_t(x)$ and $e_t^\pi(x) = J_t^\pi(x) - h_t(x)$. Their exact recursions are

$$e_t^*(x) = \inf_{a \in A_t(x)} \{A_t^0(x, a) + \beta_t P_t^a e_{t+1}^*(x)\},$$

$$e_t^\pi(x) = A_t^0(x, \pi_t(x)) + \beta_t P_t^{\pi_t(x)} e_{t+1}^\pi(x), \quad e_T^* = e_T^\pi = 0.$$

The reference need not be fitted, and a numerical routine need not evaluate h_t separately if it encloses its signed advantages directly from the primitives.

COROLLARY 24—Reference-centered Bellman certificate: *Apply the lower-cover and robust upper-candidate construction of Theorem 23 to the excess recursions above, using verified enclosures of A_t^0 and the original probability kernels. Denote the resulting piecewise excess bounds by ℓ_t and u_t . Then, at every true state,*

$$\ell_t(x) \leq J_t^*(x) - h_t(x) \leq J_t^\pi(x) - h_t(x) \leq u_t(x).$$

Consequently $\sup_x \{u_t(x) - \ell_t(x)\}$ is an all-state loss certificate for the original constructed policy against the original continuous-action Bellman optimum. There is no separate reference-approximation allowance when the advantage enclosure is obtained directly from the exact reference law.

For an independently fixed feasible policy $\hat{\pi}$, an upper-only excess recursion using its full acquired-actor action range gives $\hat{u}_t \geq J_t^{\hat{\pi}} - h_t$. The quantity $\sup_x \{\hat{u}_t(x) - \ell_t(x)\}$ therefore certifies that particular returned policy. It does not require the policy to be regenerated on the verification grid.

PROOF: Subtract h_t from the Bellman equation and from the policy-value equation to obtain the two recursions. The induction in Theorem 23 applies verbatim to these transformed equations: kernel monotonicity and the complete feasible-action

cover give the lower excess bound, and implementability and the upper continuation bound give the upper excess bound. Both endpoints at a given true state contain the identical $h_t(x)$, which cancels in their difference. The same upper induction applies to any fixed feasible actor when its complete action range is enclosed on every verification cell. Subtracting the lower optimum bound proves the final assertion. *Q.E.D.*

In the two-date investment economy, choose the original zero-investment policy as reference. Its final-date advantage is the original analytic final-cost difference. At the initial date, use a common innovation to propagate the state difference caused by the proposed action and integrate the difference of subsequent zero-policy costs. Centered polynomial identities evaluate that difference; the final innovation's common variance cancels, while its correct marginal law is retained. The construction therefore does not assume a policy-value oracle, a differentiable acquired policy, or a learned invariant.

The centered experiment is explicitly subsequent to the uncentered one. It retains the unsuccessful original tolerance tests and applies a separately frozen five-rung verification design. The same centered lower table can also certify the previously returned learned and conventional policies. Such revalidation has its own cost and does not retroactively alter the original prospective stopping criterion or its measured clock. This connects the original Bellman-accuracy question to the actual returned objects, rather than only to an alternative constructed policy.

11. STABLE ENCLOSURES AND CERTIFICATION OF A RETURNED POLICY

A valid one-step interval and a stable recursion are different requirements. The first establishes a runtime bracket; the second controls how an approximation allowance propagates when the next-date input changes. We make the distinction explicit and then apply the common-reference construction to a policy that has already been returned by a prospective service.

11.1. Fixed-cover stability

At a fixed approximation level, hold the state cells, action covers, candidate actions, innovation bins and their probabilities fixed. An endpoint map may take minima or maxima over the cells intersecting a transition box, average those endpoints with nonnegative weights of total mass one, add a stage-cost endpoint and finally minimize over a fixed finite family of action intervals or feasible candidates. A reference-shifted endpoint map may additionally add a fixed signed reference advantage. These operations describe the exact endpoint maps beneath the outward arithmetic used here.

LEMMA 25—Stability of the executable endpoint maps: *Every such exact endpoint map $\mathcal{E}_t$ is order preserving and satisfies*

$$\|\mathcal{E}_t f - \mathcal{E}_t g\|_\infty \leq \beta_t \|f - g\|_\infty. \quad (49)$$

Before intersection with an independent bound, it also satisfies $\mathcal{E}_t(f + c) = \mathcal{E}_t f + \beta_t c$ for constants c . Intersecting a lower endpoint with an independently valid lower bound by taking a maximum, or an upper endpoint with an independently valid upper bound by taking a minimum, preserves order and the nonexpansive estimate when the independent bound is fixed. A numerical endpoint perturbation with verified uniform magnitude δ_t enters the propagated error as an additional local allowance, rather than being presumed nonexpansive because it is outward.

PROOF: For any nonempty finite index set I , both $\min_{i \in I} f_i$ and $\max_{i \in I} f_i$ are order preserving and one-Lipschitz in the supremum norm. A nonnegative weighted average of total mass one has the same properties. A fixed additive term does not alter a difference, and multiplication by $\beta_t \geq 0$ supplies the factor in (49). The final minimum over actions is again one-Lipschitz. Each of these operations commutes with addition of a constant in the stated way. Finally, $x \mapsto \min\{x, c\}$ and $x \mapsto \max\{x, c\}$ are monotone and one-Lipschitz. A separate numerical perturbation is bounded by the triangle inequality. Q.E.D.

Consequently, the consistency clause of Theorem 23 applies to the fixed-cover maps actually used in the computation. If the ideal lower and upper maps at the

true next Bellman value have errors at most α_t , and their numerical evaluation has uniformly bounded endpoint error δ_t on the relevant range of inputs, put $\eta_t = \alpha_t + \delta_t$. Then the two one-sided errors satisfy

$$e_t^- \leq \eta_t + \beta_t e_{t+1}^-, \quad e_t^+ \leq \eta_t + \beta_t e_{t+1}^+.$$

Their sum gives the displayed two-sided finite-horizon bound. Different approximation levels may use different covers; the stability comparison is within each fixed level. Arbitrarily small tolerances require all nonvanishing sources of error to be refined, including the action quantum when it is a binding discretization. A finite ladder at quantum $1/4096$ is not, by itself, an executed zero-error limit.

Inclusion alone would not justify this step. In a one-state example with continuation $v \geq 0$ and discount β , the endpoint $\beta v + v^2$ is a valid upper bound on βv and is exact at $v = 0$. Its variation from 0 to 1 nevertheless exceeds β . Thus an ideal error statement only at the true continuation cannot replace (49). The present proof supplies the needed property for the actual min-max and probability-average maps; it does not assume stability of every conceivable enclosing procedure.

11.2. *The original optimum and the actual returned actor*

Let $h_t = J_t^0$ denote the value of the installed zero-investment policy in the original model. A common-reference Bellman calculation supplies a function ℓ_t satisfying

$$\ell_t(x) \leq J_t^*(x) - h_t(x).$$

Now fix a complete, previously returned acquired policy π . Its observation partition and quantum action indices are part of its identity and are not changed by verification. On a verification cell C , let $R_t(C)$ contain every action actually taken by π_t at a state in C . Such a rectangle is an outer enclosure of the graph of the fixed policy, not a proposal to implement every action in that rectangle. In particular, its endpoints need not be feasible throughout C ; the actual policy must remain feasible at each actual state, and the primitive formulas used on the outer enclosure must remain valid there.

Suppose $\mathcal{V}_t(C)$ is an outward enclosure, for all actual graph points $(x, \pi_t(x))$ with $x \in C$, of

$$A_t^0(x, \pi_t(x)) + \beta_t P_t^{\pi_t(x)} u_{t+1}(x), \quad (50)$$

where $A_t^0(x, a) = c_t(x, a) + \beta_t P_t^a h_{t+1}(x) - h_t(x)$ and $u_T = 0$. Define $\tilde{u}_t(C)$ as the upper endpoint of $\mathcal{V}_t(C)$.

PROPOSITION 26—Paid revalidation of an unchanged policy: *If the stored actor is feasible and the enclosures in (50) include every reached state and the original innovation law, the recursion $u_t = \tilde{u}_t$ gives*

$$\ell_t(x) \leq J_t^*(x) - h_t(x) \leq J_t^\pi(x) - h_t(x) \leq u_t(x).$$

If an independent whole-domain policy certificate has established $J_t^\pi \leq h_t$ at every date, the recursion may instead use

$$u_t(C) = \min\{\tilde{u}_t(C), 0\}. \quad (51)$$

In either case, an outward maximum of $u_t(C) - \ell_t(C)$ bounds the loss of this particular returned actor against the original continuous-action Bellman optimum. Verification does not change the actor or require a policy-cost sampling event.

PROOF: The identity

$$J_t^\pi - h_t = A_t^0(\cdot, \pi_t) + \beta_t P_t^{\pi_t} (J_{t+1}^\pi - h_{t+1})$$

holds under the original kernels. Starting from the common terminal value, monotonicity of expectation and the whole-graph enclosure prove the upper inequality by backward induction. The lower inequality is the previously established Bellman lower bound; feasibility gives the middle inequality. When $J_t^\pi - h_t \leq 0$ has been independently proved, intersection with zero preserves the upper bound at each induction step. Subtracting the two bounds cancels the same $h_t(x)$ pointwise. No difference between extrema at unrelated states is substituted for this cancellation. Q.E.D.

The independent property used in (51) is obtained here by reconstructing the original reference-advantage gate at every acquired cell and date. Its nonpositive upper endpoints imply $A_t^0(x, \pi_t(x)) \leq 0$ throughout the domain, and backward policy improvement gives $J_t^\pi \leq J_t^0$. The revalidation also retains the entirely unclipped upper recursion and its final gaps. This permits the reader to distinguish a justified independent bound from a numerically favorable truncation. A record of nonpositive values at sampled centers would not supply the required property.

The verification grid is finer than the actor’s acquisition partition. Actor ranges include all leaves intersecting each closed verification cell, including both sides of a discontinuity. Rectangle extrema and original-law innovation bins then enclose the continuation. These calculations have a real cost: the common lower-table construction, each fixed-actor upper recursion and the whole-process return overhead are separate measured services. They do not retroactively reduce a prospective service’s original stopping cost or certify an unexecuted high-dimensional case.

12. ORIGINAL-OPTIMUM ACCURACY IN THE INVESTMENT ECONOMY

The exact witness theorem answers an implementation question conditional on a continuation. An economist also needs to know how close the complete implemented policy is to the original optimum. We now establish that connection in the investment economy itself. The argument exploits a property not used by the earlier cell-range calculation: despite the bilinear transition, the Bellman objective remains jointly convex. This permits a second-order interpolation account and transfers verified nodal decisions to an implementable policy over the entire state domain. It does not assume that a fitted neural continuation is convex.

Relation to the preceding two-control query. The earlier query experiment used the fixed shared capacity $1/8$. The complete dynamic extension below uses the original state-dependent capacity $c(x)$, with the retained exposure columns, action-cost interaction and second shock. Its nesting claim concerns the original scalar economic model; it is not an assertion that a full dynamic policy was already present in the earlier isolated query.

12.1. Primitives and Bellman regularity

Let $d \geq 2$ be even, let indices be cyclic, and set $\beta = 15/16$. Write

$$S(x) = \frac{2}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2 \left(\frac{1}{2} - \frac{2}{d} \sum_j x_j \right)^2,$$

$$g(x) = S(x) + \frac{2}{d} \sum_j (x_j - 5/8)^2, \quad c(x) = \frac{1}{8} + \frac{1}{8d} \sum_j x_j.$$

The original scalar control has $0 \leq a \leq c(x)$, running cost $S(x) + a^2 + 4a^4$, and transition

$$F_j(x, a, z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + B_j a + s_j z,$$

where $(B_j, s_j) = (1/2, 1)$ at alternating coordinates and $(1/4, -1)$ at the others, and z is uniform on $[-1/32, 1/32]$.

For the retained two-control extension, $a_1, a_2 \geq 0$, $a_1 + a_2 \leq c(x)$, the second exposure column interchanges $1/2$ and $1/4$, and the action cost is

$$R(a) = a_1^2 + a_2^2 + 4(a_1^4 + a_2^4) + a_1 a_2 / 4.$$

An independent common-direction uniform innovation of radius $1/64$ is added. This is the previously specified extension, nested at $a_2 = 0$ when that innovation is disabled. Both models preserve $[0, 1]^d$. Let $m \in \{1, 2\}$ denote the number of controls and define

$$Q_t^*(x, a) = S(x) + R(a) + \beta \mathbb{E} V_{t+1}^*(F(x, a, Z)), \quad V_t^*(x) = \min_{a \in A(x)} Q_t^*(x, a), \quad V_T^* = g.$$

All costs and laws in this display are the economic primitives, not a fitted surrogate.

THEOREM 27—Regularity of the original Bellman problem: *For either control specification and every finite horizon, V_t^* is convex and continuously differentiable in the interior, with the corresponding one-sided extension on the cube. It is G_t -Lipschitz in the ℓ^1 norm and M_t -semiconcave in the Euclidean norm, where valid constants are*

$$G_T = 10/d, \quad G_t = \frac{1037}{128d} + \beta \frac{47}{64} G_{t+1},$$

$$M_T = 26/d, \quad M_t = \frac{22}{d} + \frac{21}{256d} + \beta \left(\frac{9}{16} M_{t+1} + \frac{1}{8} G_{t+1} \right).$$

In particular, $G_t \leq 27/d$ and $M_t \leq 54/d$. The function Q_t^* is jointly strongly convex on its convex feasible graph, with state curvature at least $107/(128d)$ and action curvature at least $7/4$.

A valid semiconcavity constant along each coordinate line is

$$D_T = \frac{9}{d} + \frac{16}{d^2},$$

$$D_t = \frac{5}{d} + \frac{16}{d^2} + \frac{21}{256d^2} + \beta M_{t+1} \left(\frac{85}{256} + \frac{23}{256d} \right).$$

Consequently, for the positive multilinear interpolant I_n on a mesh of spacing $1/n$,

$$0 \leq I_n V_t^*(x) - V_t^*(x) \leq \kappa_t, \quad \kappa_t = \frac{dD_t}{8n^2}. \quad (52)$$

PROOF: The state cost has gradient infinity norm at most $8/d$, Hessian norm at most $22/d$, and diagonal second derivatives at most $5/d + 16/d^2$. For the terminal cost the corresponding bounds are $10/d$, $26/d$, and $9/d + 16/d^2$. They follow by differentiating the quadratic terms and the continuously differentiable squared shortage. The bound on the cycle Laplacian is four. The shortage Hessian, where it exists, is $16\mathbf{1}\mathbf{1}^\top/d^2$; the same second-difference bound holds across its kink.

For feasible (x, a) and (y, b) , put $w = x - y$. The difference between the transition at their convex combination and the corresponding combination of transitions is $\theta(1 - \theta)w_j w_{j+1}/16$ in coordinate j . Convexity and the G_{t+1} Lipschitz bound therefore give a convexity defect of at most

$$\frac{\beta G_{t+1}}{16} \theta(1 - \theta) \sum_j |w_j w_{j+1}| \leq \frac{\beta G_{t+1}}{16} \theta(1 - \theta) \|w\|_2^2.$$

The state cost supplies curvature $4/d$, while the two-control action cost has Hessian eigenvalues at least $7/4$. Thus the state curvature of Q_t^* is at least $4/d - \beta G_{t+1}/8$, which is at least $107/(128d)$ when $G_{t+1} \leq 27/d$. Partial minimization over the affine capacity graph preserves the state convexity inequality. The minimizing action is unique, including at a binding capacity constraint.

To propagate Lipschitz continuity, write $a = c(x)u$, where u lies in the fixed unit simplex. Transfer the same u from x to y . Each action-cost partial derivative has absolute value at most $13/16$, and $\sum_j u_j \leq 1$. The action-cost transfer contributes at most $13\|x - y\|_1 / (128d)$. The transition Jacobian at a fixed action has column sums at most $11/16$; the capacity transfer adds at most $3/64$. Comparing a minimizing normalized action in both directions gives the stated recursion for G_t . Its right side maps $27/d$ below itself.

For a fixed u , the transition as a function of x has Jacobian norm at most $11/16 + 1/16 = 3/4$. Its second-order composition term is bounded by $G_{t+1}/8$. The action-cost Hessian has norm at most $21/4$, and the capacity gradient has squared norm $1/(64d)$. These facts give the recursion for M_t . The infimum over u preserves this common semiconcavity bound: after subtracting the same quadratic, each fixed- u function is concave, and an infimum of concave functions is concave. The right side maps $54/d$ below itself. Convexity together with semiconcavity yields the claimed differentiability and gradient regularity; this conclusion does not require a differentiable optimizing-action map.

Along one state coordinate, the fixed- u transition has zero second derivative. The squared norm of its first derivative is at most $85/256 + 23/(256d)$: the two original nonzero entries are bounded by $9/16$ and $1/8$, and the capacity-induced entries are bounded by $1/(16d)$. Combining this with the diagonal stage-cost bound gives D_t . Finally, positive interpolation along one coordinate has error between zero and $D_t h^2 / 8$. Applying the coordinate interpolations successively and adding the bounds proves (52). *Q.E.D.*

The theorem is a structural statement about these economic primitives. It is not a claim that every controlled diffusion, recursive preference specification, or game in the complete development edition has a convex Bellman problem. Nor does it require a signed-weight neural fit to preserve convexity. Convex approximation of dynamic programs has an established literature; the additional work here is to verify the needed curvature for the original nonlinear capacity-constrained model and connect it to the actual NBO actor and complete error account.

12.2. *Certified nodal Bellman bounds*

Suppose nodal arrays satisfy $l_{t+1} \leq V_{t+1}^* \leq u_{t+1}$ at all grid vertices. Equation (52) gives the functions $I_n l_{t+1} - \kappa_{t+1}$ and $I_n u_{t+1}$ as valid lower and upper continuations. The lower function need not itself be convex; convexity is used for the unknown true value and for the comparison argument, not asserted of arbitrary endpoint arrays.

Divide each uniform innovation component into q equal bins and use its conditional mean. Convexity gives a lower quadrature bound for the true continuation. Semiconcavity gives an upper error

$$\zeta_{t+1} = \frac{M_{t+1}}{6q^2} \sum_{r=1}^s \rho_r^2 \|s_r\|_2^2, \quad (53)$$

where $s = 1$ or 2 , $\rho_1 = 1/32$, and $\rho_2 = 1/64$ in the extension. The expectation remains the original continuous law. At the terminal date g is evaluated directly, so no terminal interpolation allowance is needed.

At a vertex v , let $\mathcal{A}_A(v)$ be the capacity simplex lattice with spacing $c(v)/A$, including all its faces. For $m = 1, 2$, a minimizer can be represented by neighboring lattice vertices whose coordinate variances sum to at most $mc(v)^2/(4A^2)$. A valid action smoothness constant is

$$H_t = \frac{21}{4} + \beta M_{t+1} \|B\|_2^2, \quad \|B\|_2^2 = \frac{5d}{32} \ (m=1), \quad \|B\|_2^2 = \frac{9d}{32} \ (m=2).$$

Taylor's inequality and that barycentric representation imply

$$0 \leq \min_{a \in \mathcal{A}_A(v)} Q_t^*(v, a) - V_t^*(v) \leq \delta_t := \frac{mH_t}{128A^2}. \quad (54)$$

The linear Taylor terms cancel even when the optimum lies on a face. This is an error against the continuous feasible optimum, not only against the finite action menu.

Let $q_t^-(v, a)$ and $q_t^+(v, a)$ include all interpolation, quadrature and arithmetic errors and enclose $Q_t^*(v, a)$. Then

$$l_t(v) = \max\{0, \min_{a \in \mathcal{A}_A(v)} q_t^-(v, a) - \delta_t\}, \quad u_t(v) = \min_{a \in \mathcal{A}_A(v)} q_t^+(v, a) \quad (55)$$

are valid nodal Bellman bounds. The zero truncation uses the nonnegative economic costs. Extra neural or conventional proposals may improve the upper side but do not remove the lower action-cover allowance. Positive interpolation, averaging and minimum maps are monotone and nonexpansive; the separate allowances in (52)–(54) are therefore propagated, rather than being discarded because an endpoint calculation contains the truth.

12.3. From a fitted witness to the complete implemented policy

Let $a_t(v) \in A(v)$ be the actual action returned at each vertex by any generator. It may be selected by the exact neural witness, a conventional fitted continuation, or their common verified fallback. Write $w_v(x)$ for the multilinear weights and set $\bar{\pi}_t(x) = \sum_v w_v(x) a_t(v)$. Affineness of capacity makes this action feasible. Let $q_t^{\text{sel},+}(v)$ be a verified upper bound on its original-optimum continuation value $Q_t^*(v, a_t(v))$, using the same lower and upper Bellman reference for all generators.

THEOREM 28—Actual-policy, whole-domain certificate: Set

$$e_t = \max_v \{q_t^{\text{sel},+}(v) - l_t(v)\} + \kappa_t.$$

Suppose the implemented policy $\hat{\pi}_t$ is feasible and differs from $\bar{\pi}_t$ by at most b_t in action ℓ^1 norm, uniformly over the state cube. Define

$$\Lambda_t = \frac{13}{16} + \beta \frac{3d}{8} G_{t+1}, \quad D_T^\pi = 0, \quad D_t^\pi = e_t + \Lambda_t b_t + \beta D_{t+1}^\pi.$$

Then, at every true state and date,

$$0 \leq J_t^{\hat{\pi}}(x) - V_t^*(x) \leq D_t^\pi.$$

This statement is conditional on the actual recorded policy, not on success of its fitting algorithm. It applies equally to a pure fitted policy and to a policy augmented by verified common candidates.

PROOF: Joint convexity, including the feasible graph, gives

$$Q_t^*(x, \bar{\pi}_t(x)) \leq \sum_v w_v(x) Q_t^*(v, a_t(v)) \leq I_n q_t^{\text{sel},+}(x).$$

The lower interpolation bound gives $V_t^*(x) \geq I_n l_t(x) - \kappa_t$. Subtraction proves the uniform optimal-advantage bound e_t . The partial action derivatives of Q_t^* are bounded by Λ_t , so feasible implementation error adds at most $\Lambda_t b_t$. Adding the future policy loss and proceeding backwards proves the recursion. No convexity of $J^{\hat{\pi}}$ is used. Q.E.D.

In the execution, observation is rounded down at forty bits. An acquired state $y \leq x$ has no greater capacity than the true state. Each interpolated action is then rounded down at twenty bits using the lower endpoint of its arithmetic enclosure. If L_{ij} bounds the derivative of action coordinate i with respect to state coordinate j , and η bounds native interpolation error, a valid implementation allowance is

$$b_t \leq 2^{-40} \sum_{i,j} L_{ij} + m(2^{-20} + 2\eta + \eta_{\text{out}}).$$

Here L_{ij} is computed from the maximum adjacent nodal difference times n , and η_{out} covers endpoint subtraction. All these numbers are recorded outward. This new barycentric acquired actor is explicitly distinguished from the earlier discontinuous cell-constant actor; an old actor is not silently replaced during a purported fixed-policy revalidation.

12.4. Arithmetic, work, and scope

The native interpolation kernel forms nonnegative tensor weights and a weighted sum of at most 2^d stored binary64 endpoints. It is compiled without reassociation or floating-point contraction. For $K = (3d + 4)2^d + 32$, an explicit conservative absolute allowance is

$$\eta = \frac{K2^{-53}}{1 - K2^{-53}} \max_v |z_v| + K2^{-1022}.$$

The final term conservatively covers underflow. Input coordinates, affine indexing, finite endpoints and the binary64 format are checked. Roundoff expansion and the ordinary interval cost arithmetic remain separate from the analytic approximation bounds. Exact rational fixtures compare the native sum with its mathematical tensor interpolant.

With $N = (n + 1)^d$ vertices, $K_A = A + 1$ scalar actions or $K_A = (A + 1)(A + 2)/2$ two-control actions, the reference calculation uses $O(TNK_A q^s 2^d d)$ elementary work under the explicit tensor implementation. Stored reference values use $O(TN)$ words and a policy uses $O(TNm)$ words. Streaming batches avoid a state-pair table. Proposal fitting, exact rational witness work, actor transfer, serialization, independent cost inference and compilation are additional costs, not absorbed into this bound. The analytic errors are second order in state, action and innovation resolution; the state-count and tensor-evaluation factors still depend exponentially on dimension. The result supplies a matched original-optimum comparison without asserting that these costs have disappeared.

13. FROM RETURNED POLICIES TO ECONOMIC RESOURCE DECISIONS

An all-state upper bound and a comparison of actual policy costs answer different economic questions. We now connect the policies returned by a construction frontier to direct cost inference and then to a choice of implementation resources. The construction data select the policies; independently generated evaluation paths assess them. This separation permits a valid comparison of first-crossing policies with different resolutions.

13.1. *Simultaneous inference after construction*

Let $\mathcal{H}$ denote all construction information: primitive inputs, numerical labels, fitted continuations, returned policies and complete certificate ladders. Conditional on $\mathcal{H}$, fix a finite catalogue of implemented-policy contrasts. Each implementation includes its observation rule, original witness or nearest-node selector, true-state feasible repair and action quantization. For a contrast i , let D_i be the difference in expected discounted costs under a specified initial law. A common-innovation path score evaluates the two policies on their own endogenous trajectories. It is not the difference between two regret bounds.

Suppose the initial and innovation uniforms are partitioned into equal bins. Independent uniform bin indices are drawn, and interval simulation encloses the score for every continuous realization inside the drawn bins. All unresolved state, selector and repair branches remain in the enclosure. If $[A_i, B_i]$ is a deterministic

support interval, clip the path endpoints to that support. Write $\underline{m}_i, \bar{m}_i$ for outward sample-mean bounds on the lower and upper endpoint samples and $v_{i,L}, v_{i,U}$ for upper bounds on their unbiased sample variances. For $n \geq 2$ define

$$R_i(v) = \sqrt{2v\ell/n} + \frac{7(B_i - A_i)\ell}{3(n-1)}, \quad \ell \geq \log(4K/\alpha), \quad (56)$$

where K bounds the number of sampled contrasts.

THEOREM 29—Construction-conditional economic intervals: *Under the stated independent-bin model, conditional on any valid construction history $\mathcal{H}$, with probability at least $1 - \alpha$ all catalogue contrasts satisfy*

$$D_i \in [l_i, u_i] := [\max\{A_i, \underline{m}_i - R_i(v_{i,L})\}, \min\{B_i, \bar{m}_i + R_i(v_{i,U})\}]. \quad (57)$$

The same coverage holds unconditionally. Subsequent selection among these covered contrasts, or evaluation at any deterministic implementation-price vector, needs no additional union bound. This conclusion applies only to policies and initial laws included in the catalogue; it does not create direct-cost observations for every unstudied tolerance or initial state.

PROOF: For each drawn bin sequence, conditional integration over its continuous realizations places the exact path expectation between the interval endpoints. Uniform mixing over the equal bins recovers the original continuous law. The lower endpoint expectation is thus no larger than D_i , and the upper endpoint expectation is no smaller. Apply the bounded empirical-Bernstein inequality of [Maurer and Pontil \(2009\)](#) to the endpoint samples and take the union bound, with outward arithmetic included in the mean, variance and radius. Conditional validity for every $\mathcal{H}$ implies unconditional validity by integration. On this one simultaneous event, all ensuing deterministic comparisons of the covered quantities hold together. *Q.E.D.*

In particular, selecting the first policy to meet a specified certificate is permitted when that selection uses construction records rather than the subsequent evaluation paths. Repeated construction clocks are not independent draws from a neural-optimizer population. The statistical randomness here belongs to the evaluation

design. A fixed pseudorandom stream implements a reproducible experiment; it is not itself a proof of the independent-bin assumption.

13.2. Replacement-fee identification

Write $D = J^W - J^F$. If witness is installed, replacing it by FVI changes total cost by $k - D$, where $k \geq 0$ is the replacement fee. If FVI is installed, the change is $k + D$. Thus the two directional gains before fees are D and $-D$.

COROLLARY 30—The break-even fee frontier: *On an event where $D \in [l, u]$, the two directional gains belong to $[l, u]$ and $[-u, -l]$. The unknown absolute cost difference satisfies*

$$|D| \in [\text{dist}(0, [l, u]), \max\{|l|, |u|\}]. \quad (58)$$

For every $k \geq \max\{0, u, -l\}$ neither replacement strictly lowers expected cost. By contrast, $k \geq \max\{0, l, -u\}$ only removes a certified strictly profitable replacement. The two thresholds generally differ. Strict inequalities in the first criterion certify that both replacements strictly increase cost.

PROOF: Subtract the gain intervals from k and compare the resulting interval endpoints with zero. Taking the image of $[l, u]$ under the absolute-value map gives (58). The assertions hold simultaneously for every nonnegative fee on the original coverage event. *Q.E.D.*

The fee frontier is more informative than a conclusion at one generous charge. The earlier 1/64 replacement decision is retained with its original interpretation and confidence account. The new analysis reports the two directional gain intervals and both thresholds instead of selecting a favorable fee after seeing the data. There is no welfare calibration hidden in the units: a decision maker must supply the economically relevant charge, or use the full frontier to see what remains unidentified.

13.3. Observation costs on the same bit grid

Let $\mathcal{B} = \{4, \dots, 16\}$ be the available bit precisions. For one fixed returned controller, write J_b for the expected cost of its actual b -bit, robustly repaired and quantized implementation. Use $b_0 = 16$ as a common reference and obtain simultaneous intervals for $D_b = J_b - J_{b_0}$. The identity interval at $b = b_0$ is exactly $[0, 0]$. Let $s_b = db \sum_{t < T} B_t$ be the discounted number of coordinate-bits purchased; the net cost is $J_b + \lambda s_b$.

THEOREM 31—Certified choice of actual net implementation cost: *Suppose all $D_b \in [l_b, u_b]$ simultaneously. For any $\lambda \geq 0$ define*

$$L(\lambda) = \min_b \{l_b + \lambda s_b\}, \quad U(\lambda) = \min_b \{u_b + \lambda s_b\}, \quad (59)$$

$$\hat{b}(\lambda) = \min \arg \min_b \{u_b + \lambda s_b\}.$$

Then, simultaneously for all $\lambda \geq 0$,

$$0 \leq J_{\hat{b}} + \lambda s_{\hat{b}} - \min_{b \in \mathcal{B}} (J_b + \lambda s_b) \leq U(\lambda) - L(\lambda). \quad (60)$$

The lower and upper functions are finite piecewise-affine envelopes. All changes in their active lines occur at nonnegative intersections of catalogue lines, with exact comparisons deciding ties. The smallest selected precision $\hat{b}(\lambda)$ is weakly decreasing in λ .

PROOF: Subtract the common J_{b_0} . The selected implementation has net difference at most $U(\lambda)$, and the best catalogue implementation has net difference at least $L(\lambda)$. This proves the regret bound pointwise on the one simultaneous event and hence for every λ . A minimum of finitely many affine functions changes its active affine expression only at an intersection. For $\lambda_2 > \lambda_1$, sum the two minimizing inequalities to obtain $(\lambda_2 - \lambda_1)(s_{\hat{b}(\lambda_2)} - s_{\hat{b}(\lambda_1)}) \leq 0$. Since s_b increases strictly with b , the precision comparative static follows. *Q.E.D.*

This result makes a statement about *actual* net cost within the declared bit catalogue. It differs from minimizing a uniform all-state loss bound. Both criteria are useful: the latter gives distribution-free protection relative to the full-information

optimum, while (60) uses a specified initial law to compare feasible implementations of one returned policy. Neither criterion gives an optimum over arbitrary sensor technologies or all policies.

For a fair comparison the conventional actor receives its own acquisition allowance. Suppose its nearest-node policy has a certified selected-policy account at a reported state, obtained from the graph modulus L_t^{gr} , and its actual interpolated critic has modulus L_t^f . Transporting the same selected node action to the true state, then using cell-wide capacity repair, adds at most

$$(L_t^{\text{gr}} + L_t^f + 2D_t K_t^A) r(b) + D_t \Delta_t. \quad (61)$$

Indeed, transport the feasible node pair directly to the true state, bound its node distance by the reported-state distance plus $r(b)$, and bound $f_t(\hat{x}) - f_t(x)$ by $L_t^f r(b)$. Robust repair adds $D_t(2K_t^A r(b) + \Delta_t)$. This proves (61) without assuming actor continuity. For a witness critic $L_t^f = L_t^{\text{gr}}$, it reduces to the preceding witness allowance. The empirical comparison therefore evaluates the all-state-account minimizer, the interval upper-net-cost selector, and a descriptive midpoint selector on the *same* integer bit and price grid. The midpoint selector is not called the true optimum.

14. PROSPECTIVE ECONOMIC STOPPING AND LEARNED STRUCTURE

A policy certificate determines whether a proposed change is safe at every state. An economic stopping rule asks when the returned complete policy has attained a specified expected-cost target under the initial law. Neither question is answered by choosing the best recorded policy after all computation has finished.

14.1. A first-attainment service

Fix an installed policy π^0 , a target $q \in (0, 1)$, a finite resource ladder, and cumulative sample sizes at each stage. At a stage, construct a critic from primitives, transfer the old policy exactly to any refined partition, and execute a verified simultaneous sweep. Freeze the resulting complete policy before drawing that stage's fresh evaluation stream. At each declared look form a confidence interval for

$$D_{k,q} = \mathbb{E} J_0^{\pi^k} - q \mathbb{E} J_0^{\pi^0}. \quad (62)$$

Return the policy at the first nonpositive upper endpoint. A strictly positive lower endpoint ends the current stage and advances to the next resource stage. An unresolved interval leads to the next declared look. Exhausting the last stage returns the safe incumbent and an explicit budget-exhaustion status. It is not a conclusion that the target is unattainable.

THEOREM 32—Validity at prospective return: *Suppose each accepted sweep satisfies Theorem 8. At each declared stage and look, assume the interval for (62) has conditional failure probability at most $\alpha_{k,l}$ given the information preceding the stage's fresh evaluation stream. If $\sum_{k,l} \alpha_{k,l} \leq \alpha$, then, with probability at least $1 - \alpha$, every successful return satisfies its stated expected-cost target. Every returned incumbent, including one returned after budget exhaustion, is statewise no worse than the original installed policy. No independence between different methods or different looks is required for the union bound. Each within-stage endpoint sample must satisfy its stated conditional sampling assumptions.*

PROOF: Condition on the history preceding each executed stage. Its frozen policy and fresh independent-bin stream satisfy the fixed-policy interval guarantee. Integrating conditional failure probabilities and taking the union over the prespecified potential looks gives a simultaneous event of probability at least $1 - \alpha$. The first-crossing rule selects one already covered interval on this event, so its nonpositive upper endpoint implies (62) is nonpositive. Unexecuted later looks need not be computed. Statewise safety follows by composition of the deterministic sweep inequalities and exact policy transfer, independently of the economic inference event. *Q.E.D.*

The theorem also permits a countable predetermined ladder with summable failure allowances. The experiment uses the finite ladder specified below, not an unimplemented anytime extension. A process clock begins before construction and ends after its durable output. Failed stages, own-policy evaluation and inference are charged to that service. A method reaching a target early must not be charged for hypothetical later stages, and a method failing early must not have its completed work omitted.

For the implemented bounded-sample account, each path is enclosed over its complete forty-bit initial and innovation cells. The last innovation is integrated analytically. The interval in Theorem 29 applies at each declared look, using the known support and outward sample moments. The new family limit is 2,048 intervals with logarithm upper bound 14 and error 1/100; the planned maximum is 1,188 intervals. Earlier looks reuse prefixes within a stage and are covered by the same finite union. Distinct stages use fresh streams. A fixed pseudorandom seed establishes reproducibility under the declared independent-bin model, not mathematical independence of a deterministic sequence.

COROLLARY 33—Economic adoption at return: *Suppose the returned policy has cumulative gain $G = \mathbb{E}J_0^{\pi^0} - \mathbb{E}J_0^{\pi} \in [l, u]$ on the simultaneous event. For every replacement charge $\tau \geq 0$, computation price $\lambda \geq 0$, and recorded complete work $W \geq 0$, its net gain belongs to*

$$[l - \tau - \lambda W, u - \tau - \lambda W].$$

Thus $\tau + \lambda W < l$ certifies profitable adoption and $\tau + \lambda W > u$ rules out profitable adoption. These comparisons hold simultaneously over the entire nonnegative price plane on the original event.

PROOF: Subtract the same known quantity $\tau + \lambda W$ from both endpoints of the gain enclosure. The assertion is pathwise on the simultaneous event, so choosing prices after observing recorded work requires no extra confidence allocation. *Q.E.D.*

This is a decision at return, not a claim that the executed relative-cost stopping rule optimizes all possible prices or future computation schedules. The units of λ must convert recorded work into the model's cost units. In an uncalibrated theoretical economy the price plane is an explicit economic sensitivity analysis, not an estimated welfare magnitude.

14.2. Learning an approximately null direction

Suppose the transition is $\mu_t(x, 0) + Ba + sZ$ with unknown $B \in [0, 1]^d$, known nonlinear drift and known bounded additive action-invariant innovation law. Fit

a direction $\hat{v}$ on training observations. Independent validation observations Y_i estimate B coordinatewise; each coordinate has mean B_j and range length at most w . Let the recorded mean enclosure be $[\underline{m}_j, \overline{m}_j]$ and let δ_{bin} cover numerical bin displacement. Form

$$\mathcal{K}_N = \prod_{j=1}^d \left[\max\{0, \underline{m}_j - r_N\}, \min\{1, \overline{m}_j + r_N\} \right], \quad r_N = w \sqrt{\ell / (2N)} + \delta_{\text{bin}}. \quad (63)$$

Only validation data, the fitted direction and the declared parameter prior enter the gate.

THEOREM 34—Confidence-qualified action-null correction: *For a finite collection of H validation boxes, each with at most $d_{\max}$ coordinates, $2Hd_{\max}e^{-\ell} \leq \alpha_B$ gives simultaneous coverage of their true exposures with probability at least $1 - \alpha_B$. On this event, for every fitted direction, amplitude M , and nonnegative feasible action pair,*

$$|(P_B^a - P_B^b)M(\hat{v}^\top y + c)_+| \leq |M| \sup_{B' \in \mathcal{K}_N} |\hat{v}^\top B'| |a - b|. \quad (64)$$

A whole-cell critic-advantage upper bound evaluated uniformly over the same box, plus the discounted right-hand side of (64), is a valid upper bound for the corresponding true advantage with the nuisance removed. Accepting only when that bound is nonpositive preserves safety simultaneously for all tested amplitudes and actions. The simplification is rejected whenever its uncertainty allowance prevents the gate from passing.

PROOF: The bounded-coordinate inequality of [Hoeffding \(1963\)](#) and a union bound give coverage of (63); numerical mean and bin allowances preserve containment. On that event, use a common innovation to couple the two actual kernels. Their next-state difference is $B(a - b)$. The positive-part map is one-Lipschitz, so its projected difference is bounded by $|\hat{v}^\top B| |a - b|$ for every realization. Integration proves (64). The box event controls all linear projections, not merely a preselected direction. Uniformly enclosing the rest of the critic advantage over that same parameter set prevents an unobserved true-parameter oracle from entering the remaining calculation. Subtracting a nuisance difference cannot increase the upper bound by more than its absolute allowance. Q.E.D.

The experiment uses $w = 1/2$, $\ell = 14$, 27 validation boxes and at most eight coordinates, giving error at most $1/200$. The direction is fitted on 256 observations and then quantized; the exact stored direction is used in (64). Validation sizes are 512, 4,096 and 32,768. The guarantee concerns an unknown action-exposure vector in a maintained structural model. It does not infer an arbitrary unknown transition law, and coverage failures must remain in the evidence rather than being discarded by reference to the simulator’s known truth.

An empty validation box invalidates the parameter qualification and rejects the simplified certificate. It cannot authorize a policy change.

15. PROSPECTIVE RETURNS IN THE NONLINEAR INVESTMENT ECONOMY

15.1. Design

Tasks are $(d, T) = (2, 2), (4, 4), (8, 6)$. Each method/task/target combination is a separate from-primitives service. Targets $q = 9/10$ and $49/50$ require ten- and two-percent expected-cost reductions relative to installed zero investment. The stopping statistic is the signed contrast $J(\pi) - qJ(0)$, not a ratio of plug-in mean estimates. Two resource stages use respectively 256 and 1,024 training states, 32 and 128 non-tensor leaves, and four and eight innovation bins. Actions are downward quantized at $1/4096$ using the lower capacity over the complete leaf. Partition geometry is chosen before the generator and is nested between stages.

Trained ReLU, quadratic and ExtraTrees fitted-value methods use their own future targets. The neural architecture and training are described in Section 6. ExtraTrees uses 16 trees, depth at most eight and minimum leaf size three; quadratic FVI fits a ridge-regularized quadratic. The original two-dimensional frontends are reconstructed at $N = 8, 16$, $K = 4$ and $M = 4$. Every method receives nine common robust candidate actions and one fitted proposal, as well as eight intervals covering the full continuous feasible action set. The new algebraic neural action solver is evaluated separately on saved critics; it was not substituted for the frozen experimental construction menu.

After one complete simultaneous sweep at a stage, inference uses cumulative looks of 4,096, 16,384 and 65,536 interval paths. It stops on the first covered target crossing, advances a stage at a strictly positive lower target contrast or exhausted

look budget, and never executes a later stage after success. All computation, including failed stages and each method’s own inference, is retained. Three isolated sequential processes repeat identical training and inference seeds, with rotated method order, a fixed available CPU core and one numerical-library thread. These measure reproducibility and clock variation, not a population of independently trained networks. Clock frequency is recorded where available but not controlled.

15.2. *The refinement and the complete returned policies*

The first cohort uses the residual/value cache. Its complete 66-process record contains 39 target attainments and 27 budget exhaustions. In the eight-dimensional task it makes no policy change. The separate sensitivity extension was motivated by that completed result and frozen before its own production and fresh inference. It retains the same targets, fitted constructions and common actions, while intersecting the old cache comparison with the signed reference interval of Proposition 10. This chronology matters: the extension is not described as prespecified before observing the primary cohort.

TABLE I

OUTCOMES BEFORE THE SIGNED REFERENCE REFINEMENT

d	T	q	Stages	Paths	Status	Last-pass changes
2	2	49/50	1	16384	Hit	27
2	2	9/10	2	65536	Hit	93
4	4	49/50	2	16384	Hit	90
4	4	9/10	2	16384	Exhausted	90
8	6	49/50	2	65536	Exhausted	0
8	6	9/10	2	4096	Exhausted	0

This primary-cache cohort is not dropped after its limitations become visible. These are the neural rows; the candidate generators return policy-identical final arrays within each primary task and target.

Exhausted means the declared finite service did not certify its target, not that improvement is impossible. All own construction, inference and failed-stage work remains charged.

All 66 signed-reference services attain their targets at the first resource stage. This means one complete all-date sweep on 32 leaves, not T exact sweeps and not

TABLE II

ACTUAL COSTS AT THE PROSPECTIVE NEURAL-POLICY RETURN

d	T	q	Paths	Expected cost	Installed minus returned
2	2	49/50	4096	[0.5338, 0.6834]	[0.0622, 0.2408]
2	2	9/10	16384	[0.5950, 0.6503]	[0.1294, 0.1777]
4	4	49/50	4096	[0.8975, 1.0658]	[0.1199, 0.3796]
4	4	9/10	4096	[0.8974, 1.0658]	[0.1177, 0.3775]
8	6	49/50	4096	[1.3427, 1.5356]	[0.0918, 0.4154]
8	6	9/10	16384	[1.4216, 1.4788]	[0.2145, 0.2978]

N is the trained ReLU generator. Every row is a separately executed target service with the signed reference certificate. Endpoints are rounded outward. The interval covers the complete implemented policy under the original continuous law. Timing repetitions have identical policy and inference identities and are not additional economic observations. Other generators and every earlier-stage outcome are retained in the supplement.

attainment of the Bellman optimum. For the neural eight-dimensional, six-date controller, the changed leaf counts by date are (26, 24, 24, 20, 14, 0). Its identified cost reduction is generated by nonterminal decisions; it is not a terminal-repair experiment relabeled as a long-horizon result. At the ten-percent target its cumulative gain interval is [0.21452658, 0.29774911], with endpoints rounded outward. The complete expected cost interval is [1.42162445, 1.47873990].

By Corollary 33, the eight-dimensional return supports every replacement-and-computation charge below its lower gain endpoint, in the model's normalized units. A charge above its upper gain endpoint rules out profitable adoption. The interval between them remains unresolved. This translates the returned policy into a full economic price sensitivity, rather than choosing a favorable fee after inspecting one estimate.

The directed comparison, all-state optimality account and initial-distribution cost interval continue to answer different questions. At date zero of the eight-dimensional neural service, the maximum candidate interval width falls from more than 14 to less than 0.610 after the signed refinement. The full-action greedy allowance remains about 1.021, and the inherited finite-sweep all-state bound is much looser than the observed expected cost gain. The experiment does not

TABLE III

WHERE THE SIGNED REFERENCE CERTIFICATE CHANGES DECISIONS

Date	Changed	Base width	Refined width	$U - C$	$C - L$	$U - L$
0	26	14.094	0.610	0.604	0.418	1.021
1	24	12.158	0.594	0.588	0.407	0.994
2	24	10.094	0.569	0.562	0.388	0.950
3	20	7.892	0.530	0.521	0.358	0.879
4	14	5.543	0.466	0.453	0.310	0.762
5	0	1.055	1.055	0.640	0.074	0.713

Eight-dimensional state, six dates, trained ReLU, first stage, 32 non-tensor leaves. Widths are maxima over candidate enclosures; decomposition entries are maxima over leaves, rounded upward. The last row uses analytic terminal differences rather than differentiating an acquired actor. A tighter comparison certificate is not itself a new policy-cost observation.

claim that 32 leaves or one sweep solve the continuous-action Bellman problem accurately at every state. It does establish the declared prospective expected-cost improvement for the complete implemented policy, together with deterministic statewise incumbent protection.

15.3. Method comparisons and complete work

The trained ReLU proposal changes no deployed cell beyond the nine common actions in either cohort. The only tenth-proposal effect is one compiled-witness cell in the two-dimensional sensitivity task, repeated across targets and timing runs. These identities are checked explicitly rather than inferred from similar cost intervals.

The conventional generators receive the same acquired states, common candidate actions, signed reference information and economic stopping rule. The audit reconstructs the trained proposal and then removes the tenth candidate from the recorded gate to determine whether that proposal changes the selected action. This is a deterministic candidate-attribution diagnostic, not a fresh policy-cost experiment. Exact deployed-policy identities are checked across generators and across timing repetitions.

TABLE IV

COMPLETE PROCESS WORK TO THE TEN-PERCENT COST TARGET

d	T	Method	Min (s)	Median (s)	Max (s)	Output KiB	RSS MiB
2	2	W	0.781	0.808	1.345	739	42.0
2	2	E	1.678	1.737	1.740	1100	110.0
2	2	Q	0.695	0.701	0.745	781	42.3
2	2	N	1.389	1.409	1.431	788	50.6
2	2	F	0.807	0.809	0.853	731	42.0
4	4	E	11.515	11.684	11.781	1286	107.3
4	4	Q	5.576	5.582	5.583	489	39.0
4	4	N	9.073	9.087	9.238	512	50.7
8	6	E	31.833	31.886	35.303	2588	112.3
8	6	Q	21.027	21.243	21.882	1350	43.8
8	6	N	26.714	26.771	30.501	1400	51.8

N: trained ReLU; Q: quadratic FVI; E: ExtraTrees FVI; W: compiled witness; F: tensor FVI. Three isolated sequential processes use identical seeds within a method. Each clock includes its own imports, construction, verification, inference and durable output. Compare methods within a task runner, not across runners. Affinity and numerical-library threads are fixed; clock frequency is not controlled. Output excludes prior-stage archives outside the service; RSS is process peak memory. Both targets and all repetitions appear in the supplement.

In the four- and eight-dimensional sensitivity tasks, the generators return policy-identical arrays within each target. In the two-dimensional task the compiled-witness return differs slightly, while the trained ReLU, quadratic, ExtraTrees and tensor-FVI returns coincide. Thus the policy improvement is not evidence that the neural generator dominates these alternatives. At the ten-percent target, quadratic FVI has the lowest recorded median complete-process work in each task. The neural backend supplies a genuinely trained nonlinear continuation and analytic integration, but those facilities must not be confused with an identified economic ranking.

The clocks include imports, primitive construction, critic fitting, acquired-policy preparation, verification, own inference, serialization and durable outputs. Method

comparisons are within each task’s controlled runner; clocks from different task runners are not a hardware-controlled scaling regression.

15.4. Neural action witnesses and learned null directions

TABLE V

EXACT LATTICE SEARCH ON SAVED FITTED NEURAL CRITICS

d	T	Full lattice	Candidates	Old index	Exact index	Critic decrease
2	2	896	10	279	303	0.0000800
4	4	768	4	191	169	0.0000289
8	6	640	6	559	555	0.0000012

One declared cell center and its saved next-date critic per task. Rational isolation and comparison agree with exhaustive rational lattice minimization. The reported decrease is in that fixed approximate Bellman objective, not actual economic policy cost. These solver regressions neither retrain a critic nor produce new policy-cost observations.

The algebraic action solver gives exact fitted-objective minimizers on the full deployed lattice in its retained regressions, using only a small subset of lattice indices. This is a constructive neural-search result. Its objective decreases concern the saved approximate continuation at a specified cell center, not the expected cost of a new deployed controller. All historical inference remains attached to the policies that were actually evaluated.

The sample-split study fits nine directions across dimensions two, four and eight and three independent direction-fitting seeds. Three validation sizes give 27 confidence boxes; three amplitudes on 64 acquired leaves give 81 cases and 5,184 cell-case decisions. All 27 recorded boxes contain the simulator exposure. The corrected gate makes 165 safe changes. No strictly harmful plug-in decision is identified in this learned-direction catalogue. That zero count is retained; it is not replaced by the large harmful count from the earlier, different known-perturbation experiment.

The primary policy-cost, sensitivity policy-cost and learned-exposure families have errors 1/100, 1/100 and 1/200, respectively. Their joint union error is at

TABLE VI
VALIDATION-QUALIFIED LEARNED DIRECTIONS

d	Validation rows	Projection-bound range	Corrected changes	Harmful plug-in
2	512	[0.0660, 0.0670]	36	0
2	4096	[0.0241, 0.0271]	54	0
2	32768	[0.0121, 0.0141]	75	0
4	512	[0.0608, 0.0636]	0	0
4	4096	[0.0217, 0.0232]	0	0
4	32768	[0.0081, 0.0102]	0	0
8	512	[0.0586, 0.0622]	0	0
8	4096	[0.0207, 0.0236]	0	0
8	32768	[0.0078, 0.0101]	0	0

Each row aggregates three independently fitted directions and three amplitudes on 64 leaves per case.

The projection range reports the minimum and maximum robust upper allowance across fitted directions, not a confidence interval for a population mean. All 27 exposure boxes cover the simulator exposure in this record. No strictly harmful plug-in choice is identified here; the earlier known-perturbation catalogue is a different diagnostic and remains unchanged.

most 1/40. This addition combines covered statements; it does not make identical policies, timing repetitions, amplitudes or reused prefixes independent observations. The learned-direction conclusion remains conditional on the known drift and maintained additive action-invariant innovation structure.

15.5. *Relation to the original evidence*

The earlier witness, FVI, acquisition, common-repair, exact-null and adaptive-grid results remain in the complete development article and its supplement. In particular, earlier adverse witness-cost comparisons are not erased by the improvement of a newly installed zero controller. The new first-attainment services replace retrospective catalogue work as the primary work-to-target endpoint for this experiment; the historical catalogue remains a valid account of a different executed procedure. The full source and result archive preserves both and states which economic policy and law each interval actually covers.

16. EXECUTING FITTED NEURAL ACTION WITNESSES

The preceding study establishes a prospective economic return and a non-tensor verification procedure. Its candidate-attribution audit also identifies a necessary next comparison: the common action menu often determines the same policy regardless of the fitted generator. The algebraic solver in Theorem 7 was tested there as a solver, not used to create the policies whose costs were reported. We now execute that solver in new prospective services and evaluate the resulting policies on new streams. The original investment laws, Bellman optimum and NBO research object are unchanged.

16.1. *A controlled action-search comparison*

The new protocol fixes the three tasks $(d, T) = (2, 2), (4, 4), (8, 6)$ before production. It compares five modes. Common-only uses the nine robust actions without fitting a critic. ReLU-menu uses a genuinely fitted 32-feature continuation and adds its 17-menu proposal. ReLU-exact retains every one of those candidates and adds the exact fitted-critic lattice minimizer. Quadratic-exact supplies the analogous exact action search for its fitted quadratic continuation. ExtraTrees-menu supplies its own future tree-based proposal. All modes receive the same exact terminal-action search, so an analytic terminal improvement is not assigned to neural representation.

The nonterminal verification rule is identical across generators: signed sensitivity of the installed zero-investment policy from Proposition 10. Its state boxes, innovation bins and primitive arithmetic do not depend on the fitted critic. Shared candidate actions therefore have identical upper endpoints. This design implements the premise of Corollary 13, rather than changing the candidate generator and its error certificate simultaneously.

Each mode is reconstructed independently for each task, target and one of three distinct training seeds. ReLU-menu and ReLU-exact use the same seed within an attempt, but neither loads the other's fit free of charge. Their fitted-parameter identities are checked. Different seeds produce different fitted neural objects. The three seeds are not a population of economic tasks, and their timing variation is not

described as a controlled timing-only repetition. The preceding fixed-seed timing repetitions remain a separate experiment.

16.2. Prospective economic targets

The two targets require ten- and twenty-percent expected-cost reductions from the installed zero policy. Each target service starts from primitives. It has two declared construction attempts, with training rows 256, 1024, acquired leaves 32, 64, and innovation bins 4, 8. The same locally split non-tensor observation geometry and downward action spacing $1/4096$ apply to every mode.

At each attempt a complete candidate is certified against the installed policy and frozen before inference. Intermediate attempts are not deployed. In particular, refinement does not apply the smooth-reference derivative formula to an already changed discontinuous actor. Every attempt is safe relative to the installed zero policy; no ordering between the policies of different attempts is assumed.

The own-service stopping estimand is $J(\pi) - qJ(0)$, with $q = 9/10$ or $4/5$, rather than a ratio of two estimated means. Its cumulative inference looks have 4096, 16384 and 65536 paths. A nonpositive upper endpoint returns immediately. A positive lower endpoint advances the construction attempt; an unresolved interval advances the declared look. After the final unsuccessful attempt the service returns a budget-exhausted safe policy, not a claim that the target is impossible. Every attempted fit, exact search, verification, inference and durable output belongs to the service's surrounding process clock.

There are 90 separately executed target services. After each set of five modes has returned, a separate 65,536-path stream compares ReLU-exact with each other returned policy. This stream is independent of construction and stopping information under the declared independent-bin model. It evaluates the original continuous initial and innovation laws using forty-bit bin enclosures and analytic final integration. Negative path differences are retained. Full policy identities receive exact zero contrasts; other overlapping intervals are not treated as identities.

The new finite confidence family has error at most $1/100$. Its maximum possible interval count is

$$90 \cdot 2 \cdot 3 \cdot 3 + 18 \cdot 4 = 1692 < 2048,$$

covering own cost, target contrast and gain at every possible look, together with the independent validation contrasts. The retained logarithm allowance is checked by exact rational arithmetic. The first-attainment argument of Theorem 32 applies conditionally at every fresh stage. This finite-look account is not an undeclared confidence sequence valid at arbitrary future sample sizes.

The new cohort executes 90 separate prospective target services. Of these, 75 attain their declared target and 15 return a budget-exhausted safe candidate. The audit regenerates 78 distinct fitted-model certificate and action-search calculations and checks 330 stopping looks. In the trained ReLU-exact services, 1354 nonterminal cell-attempt records have strictly positive menu optimization loss, while the additional algebraic candidate changes 9 deployed cell-attempt decisions. These counts refer to the declared service records, not independent samples. Among the 72 fresh paired comparisons with ReLU-exact, 0 identify lower ReLU-exact cost, 0 identify higher cost, 42 are exact policy identities and 30 remain unresolved.

16.3. *What the neural action computation changes*

Table VIII separates positive center-objective optimization loss, strict improvement of a whole-cell upper endpoint and an actual change of the deployed action. These events are not interchangeable. The center optimization loss is evaluated exactly from the same stored critic; it is zero for the algebraic minimizer. Removing only the added algebraic candidate reconstructs the within-attempt menu policy with the same verifier. Thus every reported stricter endpoint is a genuine nested-menu improvement.

The new execution supplies the previously missing connection between the trained representation’s exact action witness and a deployed policy. It also measures the extent of that connection. A large number of positive optimization losses can coexist with few additional deployed actions because the common safe menu already determines most selected upper bounds. Even a strictly improved upper

TABLE VII
PROSPECTIVE TWENTY-PERCENT TARGET SERVICES

d/T	Mode	Reached	Cost upper	Median (s)	Range (s)
2/2	C	3/3	0.61949	3.776	3.763–3.793
2/2	N	3/3	0.61949	4.511	4.497–4.536
2/2	X	3/3	0.61948	5.216	5.179–5.238
2/2	Q	3/3	0.61949	4.083	4.043–4.107
2/2	E	3/3	0.61962	5.185	5.168–5.202
4/4	C	3/3	0.97342	12.881	12.439–12.983
4/4	N	3/3	0.97342	15.077	15.004–15.240
4/4	X	3/3	0.97342	17.951	17.786–18.369
4/4	Q	3/3	0.97342	13.780	13.475–13.991
4/4	E	3/3	0.97342	15.984	15.872–16.513
8/6	C	0/3	1.36544	35.153	35.049–35.571
8/6	N	0/3	1.36544	39.239	39.190–39.458
8/6	X	0/3	1.36544	44.189	44.183–45.418
8/6	Q	0/3	1.36544	38.682	38.257–38.954
8/6	E	0/3	1.36544	40.766	40.704–41.016

C uses common candidates without fitting. N uses the trained ReLU menu; X adds its exact algebraic witness; Q uses quadratic exact search; E uses ExtraTrees menu proposals. All modes receive the same exact terminal action. Three different training seeds are reconstructed separately. Cost upper is the largest returned-policy upper endpoint across those seeds, not a confidence bound for a population mean. Complete process times include every own failed attempt and inference look.

bound does not prove that the resulting policy costs less than its menu counterpart. The independent paired contrasts in Table IX, and not the fitted objective or interval width, determine that economic comparison. Every individual interval is reported in the supplement.

The common-only control pays no fitting cost and is a substantive comparator. Quadratic-exact receives the same algebraic-search opportunity, while ExtraTrees retains its declared fitted-value construction. This prevents the exact solver’s availability from being conflated with a special verification privilege. The experiment compares the complete returned procedures and identifies their components; it

TABLE VIII
OBJECTIVE SEARCH AND WHOLE-CELL DEPLOYMENT ATTRIBUTION

d	Mode	$\kappa > 0$	Changed	Stricter	Roots	Values
2	X	163	7	7	337	3632
2	Q	145	0	0	321	2276
4	X	451	2	2	626	10849
4	Q	224	0	0	409	3965
8	X	740	0	0	840	14603
8	Q	37	0	0	127	4866

Counts are cell-attempt records across all own target services, including repeated reconstruction at the other target; they are not independent economic observations. Positive optimization loss refers to the nonterminal menu witness compared with the exact lattice witness. Changed and stricter counts remove only the extra algebraic proposal while retaining the shared verifier. Root and value counts include the common terminal solver, which is not credited as a neural-specific contribution.

does not presume that fitting a neural continuation must beat a common-action or conventional-value alternative.

16.4. *Work, uniform accuracy and adoption*

The main table reports complete own-target process work at a matched economic target. It includes failed intermediate attempts and all inference actually needed to return. The supplementary work ledger separately records fitting, root isolation, candidate evaluation, signed verification, path evaluation, serialized bytes and resident memory. Different tasks run on different task runners; within each task processes are sequential, use one available core and one numerical-library thread, and rotate mode order. Available governor information is recorded, but clock frequency is not controlled.

The fresh paired validation is additional study work. Its recorded timer ends after endpoint evaluation and serialization, before the final comparison JSON is committed. It is therefore reported as a partial validation-service clock, not a complete process clock, and is not concealed inside the own-target time. The protocol's intended complete comparison-process timing was not implemented; this timing

TABLE IX
FRESH RETURNED-POLICY COST COMPARISONS

d	X versus	X lower	X higher	Identity	Unresolved
2	C	0	0	2	4
2	N	0	0	2	4
2	Q	0	0	2	4
2	E	0	0	2	4
4	C	0	0	3	3
4	N	0	0	4	2
4	Q	0	0	3	3
4	E	0	0	2	4
8	C	0	0	6	0
8	N	0	0	6	0
8	Q	0	0	6	0
8	E	0	0	4	2

Each row covers both targets and all three training seeds. A negative paired difference favors X. Identity means equal complete policy arrays on the same observation partition, not merely overlapping intervals. Other unresolved intervals are not equivalence findings. The fresh comparison stream is independent of the corresponding stopping information under the sampling model.

boundary does not change the prospective stopping rule or its fully recorded own-service work. The raw timing records are retained without retrospective adjustment.

Every returned policy also receives a one-sweep all-state loss account. At date t it includes the old-reference continuation term $\beta_t E_{t+1}^0$ as well as the directed greedy allowance. The tables use a valid support bound for that reference error and cap the result by the original total-cost support. They do not substitute the local $U - L$ gap for loss against the Bellman optimum. A successful expected-cost target under the initial law is thus not mislabeled as exact Bellman convergence.

The gain from replacing installed zero investment remains an actual economic gain, subject to the stated confidence event and normalized primitives. Choosing between two returned methods requires the additional incremental-cost calculation (36). It charges the measured work difference and any replacement fee, and uses

the fresh paired policy-cost interval. The complete fee-computation-price regions follow from that interval without choosing a favorable fee after seeing the outcome. The experiment is a theoretical investment exercise, not an externally calibrated welfare estimate.

17. EQUAL-POLICY WORK AT PROSPECTIVE ECONOMIC TARGETS

The preceding experiment asks whether an exact trained-neural witness changes a returned policy. A different, necessary question is how economically that same witness can be computed. The lossless screen permits a controlled comparison in which economic accuracy, fitted parameters, whole-cell verification and statistical stopping remain identical. We integrate the separately frozen R58 execution into this revision and independently reconstruct its solver, certificate and path calculations. The integration creates no new training run or independent statistical observation.

17.1. *A matched intervention within Neural Bellman Operators*

The experiment retains the original nonlinear cyclic investment laws. Its tasks are $(d, T) = (2, 2), (4, 4), (8, 6)$ and its targets are ten- and twenty-percent expected-cost reductions relative to installed zero investment. Each task-target has two solver modes and three sequential complete-process repetitions, for 36 services. Both modes independently train the same 32-feature own-future ReLU continuation. Exact stored-parameter hashes establish matching; the common seed alone is not used as proof of matching.

The sole intervention is the nonterminal fitted-action solver. Algebraic search partitions the hinge objective and isolates its cubic stationary roots. Screened search performs exact activation-region reduction, outward lattice elimination and rational comparison of survivors. Both use the same robust action spacing $1/4096$, the same smallest-index tie rule and the same exact terminal solver. They retain the nine common actions and the critic-menu proposal; the exact action is an additional candidate. The same signed installed-reference verifier supplies all whole-cell upper endpoints and continuous-action lower covers.

The acquired observation geometry is the existing locally split binary partition, not a tensor grid. The two resource attempts use $(256, 32, 4)$ and $(1024, 64, 8)$ for

training rows, acquired leaves and innovation bins. All attempts are checked against installed zero investment. Intermediate candidates are not deployed, and no derivative argument is applied to a changed discontinuous acquired actor. This experiment is a prospective refinement service, not a claim that two refinement attempts are two policy-iteration sweeps.

17.2. First attainment, including nonattainment

At each attempted resolution the complete policy is frozen before inference. The target estimand is

$$D_q = J(\pi) - qJ(0), \quad q \in \{9/10, 4/5\}.$$

Cumulative sample sizes are 4096, 16384 and 65536. A nonpositive upper endpoint of D_q returns immediately. A strictly positive lower endpoint advances the construction attempt. An unresolved interval advances the inference look; an exhausted look budget advances the attempt. Failure to certify after the last attempt returns a safe policy with status *budget exhausted*. The protocol specifies these rules before its production execution, rather than selecting a favorable pass from a completed catalogue.

The R58 stopping stream is distinct from the R57 stopping and comparison streams. Within a task, target and attempt, the two solver modes and timing repetitions deliberately regenerate the same stream. Repeated computation of that stream does not multiply sample size. The confidence family has error at most 1/100 under the maintained independent-bin model. Its conservative maximum possible number of interval evaluations is

$$36 \cdot 2 \cdot 3 \cdot 3 = 648 < 2048,$$

including cost, target contrast and gain at every possible look. The frozen empirical-Bernstein implementation retains its verified logarithm allowance. The union bound requires no independence between policies or looks. It does not authorize unplanned sample sizes or an arbitrary future stopping rule.

TABLE X

COMPLETE SERVICE WORK FOR EXACTLY MATCHING POLICIES

(d, T)	Reduction	Return	Algebraic (s)	Screened (s)	Ratio
(2,2)	10%	Met	1.212	1.101	1.101
(2,2)	20%	Met	3.577	3.409	1.049
(4,4)	10%	Met	5.837	5.487	1.064
(4,4)	20%	Met	20.554	19.373	1.061
(8,6)	10%	Met	20.229	19.270	1.050
(8,6)	20%	Exhausted	63.827	62.075	1.028

Each time is the median of all three separately reconstructed fixed-work process repetitions. The ratio is algebraic divided by screened. Corresponding policies, fitted parameters, verification endpoints and stopping histories agree exactly. Exhaustion is not successful target attainment. Different tasks use different runners; the comparison is within task and target.

TABLE XI

ORIGINAL-LAW ECONOMIC COSTS AT THE COMMON RETURN

(d, T)	Reduction	Policy cost	Gain from installed	Target contrast
(2,2)	10%	[0.5964, 0.6517]	[0.1302, 0.1786]	[-0.0983, -0.0548]
(2,2)	20%	[0.5984, 0.6203]	[0.1595, 0.1740]	[-0.0171, -0.0060]
(4,4)	10%	[0.8995, 1.0672]	[0.1205, 0.3800]	[-0.2482, -0.0056]
(4,4)	20%	[0.9527, 0.9718]	[0.2616, 0.2809]	[-0.0324, -0.0167]
(8,6)	10%	[1.4177, 1.4748]	[0.2130, 0.2963]	[-0.1235, -0.0456]
(8,6)	20%	[1.3464, 1.3649]	[0.3334, 0.3558]	[-0.0141, 0.0050]

The two modes have identical intervals, not independent estimates. Endpoints are rounded outward.

The signed target contrast, rather than a ratio of estimated means, determines attainment. The twenty-percent target in eight dimensions remains unresolved at the final budget.

Thirty services attain their target. Six exhaust their budgets: the two modes, each repeated three times, at $(d, T) = (8, 6)$ and the twenty-percent target. Their final target-contrast interval contains zero. The returned policy nevertheless has a positive gain interval relative to installation. Thus failure to certify the stronger

target is not failure of policy safety, evidence of zero gain, or a proof that the economic target is impossible.

Every matched pair has identical fitted parameters, acquired actions, certificates, original-law path endpoints and stopping decisions. Consequently its policy-cost difference is exactly zero under every initial law, by Theorem 17. No separate paired statistical experiment is needed to establish this equality. In contrast, the distinct neural and conventional policies in Section 16 retain their own paired cost intervals and must not be assigned this identity conclusion.

17.3. What the computational intervention removes

TABLE XII
NONTERMINAL NEURAL SEARCH WORK, INCLUDING REJECTED LATTICE POINTS

(d, T)	Reduction	Queries	Exact A	Exact S	Roots A	Screened points
(2,2)	10%	32	189	32	12	23040
(2,2)	20%	96	612	96	43	70144
(4,4)	10%	96	723	96	41	62976
(4,4)	20%	288	2283	288	112	192000
(8,6)	10%	160	1033	160	27	94720
(8,6)	20%	480	3467	480	126	289280

A denotes algebraic search and S screened search. Counts sum all attempted resolutions in one service; all three repetitions reproduce them. The screened mode retains one point per nonterminal neural query and performs zero root isolations and zero fallbacks in this catalogue. Terminal work is common and excluded from these neural counts.

The machine-independent records identify the mechanism. Across the frozen workloads, the screen removes noncrossing integrated ridges exactly and retains one lattice point per nonterminal neural query. That point is compared rationally against the original objective. No production call uses the algebraic fallback. The number of original-objective rational evaluations therefore falls to the number of neural queries, whereas the algebraic mode compares several candidate points

per query and performs cubic-root isolation. The table charges all lattice points evaluated by the interval screen; it does not count only survivors as total work.

The screen is not an approximate optimizer with a favorable tolerance. The audit recomputes both solvers from every distinct saved model, compares every recorded witness and endpoint hash, regenerates each whole-cell certificate, and reconstructs the original interval paths. Distinct computations are memoized only after their input identities are checked. Every service file and stopping transcript is still compared. Separate exact fixtures verify all lattice endpoints for disjoint small problems, signed weights, activation boundaries, ties, non-dyadic lattices and fallback cases.

Median action-search times decrease in all six task–target groups, as do the complete process medians. The full-service algebraic-to-screened ratios range from approximately 1.028 to 1.101. These gains are smaller than the action-search gains because training, common terminal search, verification and inference are unchanged. The eight-dimensional unsuccessful target is included in the work comparison: faster exhaustion is not described as faster successful attainment.

The effect is a measured implementation gain within the trained NBO procedure at exactly matching policies and statistical accuracy. It is not a ranking of neural methods above common-only, quadratic or tree-based alternatives. The stronger conventional comparisons, the common-menu attribution results and all unresolved differences in the preceding studies remain part of the evidence.

17.4. *Complete work and the economic decision*

Every displayed service clock surrounds a separate process from invocation through exit and log flush. It includes imports, source checks, independently repeated fitting, action search, signed verification, own path evaluation and inference, all attempted resolutions and durable records. Available frequency-governor information, processor affinity, numerical-library versions, output bytes and peak resident memory are retained. The parent driver’s own checkpoint-writing overhead is outside that measured interval and is identified as controller overhead, not assigned a zero cost.

Within a task, processes run sequentially on one available core with one numerical-library thread, and solver order rotates by repetition and target. All three measurements are printed in the supplement. Frequencies are observed where available but not controlled. Different task runners do not supply a hardware-controlled dimension experiment. Three fixed-work repetitions establish neither a distribution-free runtime advantage nor a population of independently fitted policies. The three distinct fitted seeds in the preceding experiment answer a different question and remain separately identified.

For a specified computation price, Corollary 18 converts an equal-policy resource difference directly into a total-cost difference. Relative to the installed controller, the same actual gain interval supports a larger region of installation fees and computation prices when the specified implementation charge is lower. This is a substantive difference in the cost of obtaining an unchanged economic answer, not a claim that two representations of the same policy produce different welfare. The model's units remain normalized; no empirical installation fee, computation price or welfare calibration is inferred from the timings.

The original accuracy objective remains visible throughout. Every returned policy retains its one-sweep all-state loss account, including the installed-reference continuation bound as well as its full-action directed gap. Lossless search changes neither that account nor the amount of acquisition information. A successful initial-law cost-reduction target is not substituted for convergence to the original Bellman optimum. The constructive resource theory, fitted-witness transfer and directed finite-sweep theorem continue to state the conditions under which that stronger objective is controlled.

18. EXACT COMPUTATION AND THE ECONOMIC ACCURACY CONTRACT

The preceding experiments establish distinct parts of the NBO procedure. We now examine how the exact implementation behaves away from one-survivor screening, what it costs inside a complete economic service, and whether the policies actually returned can be certified against the original Bellman optimum. The original nonlinear investment family and its continuous laws are retained. The additional

two-control calculation is identified explicitly as a nested extension rather than being counted as another scalar-action policy service.

18.1. *Design, information sets, and accounting*

The completed comparison has four generators: a common-action controller with no fitted continuation, a quadratic continuation with native exact witnesses, a trained ReLU continuation with lossless screening, and the same trained ReLU construction with native finite-difference witnesses. The complete services use $(d, T) = (2, 2), (4, 4), (8, 6)$, training seeds 60103 and 60209, and targets $q = 9/10$ and $q = 4/5$. The target requires expected policy cost at most q times the expected cost of installed zero investment under the declared original initial law. These targets are not a claim about the optimum at every state.

Every service starts from the primitives. Its declared construction ladder uses $(256, 32, 4)$ and $(1024, 64, 8)$ for training samples, acquired leaves and innovation bins. The original action quantum is $1/4096$. Verification uses the same signed reference-advantage calculation, complete-cell feasibility rule and continuous-action lower cover for all four generators. Sequential inference examines 4096, 16,384 and 65,536 original-law interval paths, stopping at the first declared condition. The current execution has two hosted-worker repetitions of every mathematical design. Repeating identical seeds, policies and paths on a second worker does not double the number of independent economic observations.

The source freeze precedes the scientific services. A first execution failed in the parent recorder after an exclusive-write file was opened a second time. Its completed child outputs and failure logs remain in the attempt archive. A separately frozen correction changes that recorder to immutable numbered checkpoints and a single final ledger; the mathematical child service, seeds, targets and stopping rules are unchanged. The full corrected catalogue is used below, with no mixture of partial first-run clocks and completed rerun clocks. The separate centered-Bellman amendment was written after the uncentered frontier was observed. The fixed-policy revalidation and native factorial protocols likewise identify which predecessor results were known before their own frozen computations.

18.2. Complete services and actual policy costs

Table XIII reports every task–target–generator group on the complete return boundary. It includes imports, fitting, the executed construction and search attempts, whole-cell verification, sequential inference, durable scientific outputs, the process log and the first durable parent receipt. Cold use additionally charges native compilation; warm use starts with the compiled binary available. The source registry gives every one of the 96 services, including the 16 exhausted budgets. In total, 80 services meet their declared economic target.

The screened and native ReLU configurations return the same fitted parameters, action witnesses, acquired policy, verification arrays, stopping stage, path counts and cost intervals. Their difference is therefore an implementation-work difference at the same mathematical answer. The full-service comparison charges the native request and response as well as the exact computation. The native query benchmark below is not substituted for this complete return boundary, and a reduction in exact comparison counts is not presumed to translate proportionally into service time.

A different generator requires a different comparison. Table XV reports native-ReLU-minus-common and native-ReLU-minus-quadratic cost intervals for each task, target and seed on a fresh final-policy comparison stream. Positive intervals identify a more costly native-ReLU policy; exact identities are reported as identities; other intervals containing zero remain unresolved. The matched native-minus-screened contrast is exactly zero because their complete actor identities coincide. Conventional methods receive the same verifier and economic criterion. Their lower construction or return work is not removed from the ranking merely because they do not use learned ReLU features.

Across the twelve mathematical designs, native ReLU and each of the common-action and quadratic comparators have seven exact policy identities and five unresolved cost differences. No remaining comparison identifies a strictly lower native-ReLU cost. The common-action service has the smallest within-worker median warm return time in every task–target group; in a group whose target is exhausted by all generators this is a statement about observed work, not the work required to

TABLE XIII
COMPLETE RETURN WORK AT THE SAME ECONOMIC TARGETS

d/T	q	Mode	Met 0/1	Warm 0 (s)	Warm 1 (s)	Cold 0 (s)	Cold 1 (s)
2/2	9/10	C	2/2	1.244	1.247	1.244	1.247
2/2	9/10	Q	2/2	1.310	1.310	8.615	6.046
2/2	9/10	S	2/2	1.562	1.573	1.562	1.573
2/2	9/10	N	2/2	1.532	1.537	8.837	6.273
2/2	4/5	C	2/2	3.792	3.793	3.792	3.793
2/2	4/5	Q	2/2	3.981	3.956	11.286	8.692
2/2	4/5	S	2/2	4.820	4.818	4.820	4.818
2/2	4/5	N	2/2	4.781	4.730	12.086	9.466
4/4	9/10	C	2/2	4.347	4.366	4.347	4.366
4/4	9/10	Q	2/2	4.738	4.712	12.044	9.448
4/4	9/10	S	2/2	5.523	5.558	5.523	5.558
4/4	9/10	N	2/2	5.455	5.464	12.760	10.200
4/4	4/5	C	2/2	15.835	15.887	15.835	15.887
4/4	4/5	Q	2/2	17.012	16.904	24.317	21.640
4/4	4/5	S	2/2	19.796	19.709	19.796	19.709
4/4	4/5	N	2/2	19.524	19.858	26.830	24.594
8/6	9/10	C	2/2	16.858	16.787	16.858	16.787
8/6	9/10	Q	2/2	18.371	18.271	25.676	23.007
8/6	9/10	S	2/2	19.675	19.631	19.675	19.631
8/6	9/10	N	2/2	19.575	19.365	26.880	24.101
8/6	4/5	C	0/0	52.701	52.132	52.701	52.132
8/6	4/5	Q	0/0	57.282	57.480	64.588	62.216
8/6	4/5	S	0/0	62.443	62.636	62.443	62.636
8/6	4/5	N	0/0	61.769	61.469	69.075	66.206

C uses the common verified action family without a fitted continuation; Q uses native quadratic witnesses; S and N use the screened and native ReLU witnesses. The target is expected policy cost at most q times zero-policy cost. Each worker has two declared seeds; Met 0/1 gives the attained counts separately, each out of two. Warm and cold entries are within-worker two-seed medians of the complete durable-return clock. Cold work additionally charges that worker's measured native compilation for Q and N. Worker repetitions are not new independent economic samples. Failed targets remain in the table.

TABLE XIV
COMPLETE-SERVICE EFFECT OF REPLACING SCREENED WITH NATIVE RELU SEARCH

d/T	q	Saving 0 (%)	Saving 1 (%)	Faster 0/1
2/2	9/10	1.95	2.29	2/2
2/2	4/5	0.81	1.83	2/2
4/4	9/10	1.24	1.68	2/2
4/4	4/5	1.37	-0.75	2/1
8/6	9/10	0.51	1.35	1/2
8/6	4/5	1.08	1.86	2/2

Savings are paired relative reductions in complete warm return time, summarized by the median of the two seeds within each worker. Negative values mean slower native service. Faster counts are out of two per worker. The exact returned policy, fitted parameters, stopping stage, inference looks and intervals coincide. These are descriptive timings, not a frequency-controlled or cross-platform speed theorem.

attain the target. These outcomes distinguish the implemented NBO contract from a claim of universal neural superiority.

The additional exact witness is available only after the common action family and the finite fitted menu have been evaluated. Comparing the selected policy immediately before and after that addition identifies the decisions attributable to the extra witness under a fixed verifier. The complete reconstruction checks every such cell, including unchanged and rejected proposals. Across the distinct reconstructed verification objects, it records 12 additional-witness cell changes. This count includes the tested fitted generators; it is not an independent sample size, and it does not imply an identified average welfare gain for each changed cell. Actual economic consequences are assessed by the full-policy intervals rather than by assigning a policy value to a local reduction in a certificate.

18.3. *A language- and representation-matched action-search experiment*

The eight-method stress experiment varies three training-style fixture seeds, two feature widths, three action-lattice cardinalities and five regimes. The regimes include mixed activation, activation crossing, large signed cancellation, an exact tie between neighboring actions and a flat objective. Every rational objective is checked

TABLE XV
ACTUAL RETURNED-POLICY COST: NATIVE RELU MINUS CONVENTIONAL COMPARATORS

d/T	q	Seed	N minus C	N minus Q
2/2	9/10	60103	[0.00000, 0.00000]	[0.00000, 0.00000]
2/2	9/10	60209	[-0.00514, 0.00515]	[-0.00514, 0.00515]
2/2	4/5	60103	[-0.00517, 0.00515]	[-0.00517, 0.00515]
2/2	4/5	60209	[-0.00545, 0.00488]	[-0.00545, 0.00488]
4/4	9/10	60103	[-0.00781, 0.00751]	[-0.00781, 0.00751]
4/4	9/10	60209	[0.00000, 0.00000]	[0.00000, 0.00000]
4/4	4/5	60103	[-0.00769, 0.00760]	[-0.00769, 0.00760]
4/4	4/5	60209	[0.00000, 0.00000]	[0.00000, 0.00000]
8/6	9/10	60103	[0.00000, 0.00000]	[0.00000, 0.00000]
8/6	9/10	60209	[0.00000, 0.00000]	[0.00000, 0.00000]
8/6	4/5	60103	[0.00000, 0.00000]	[0.00000, 0.00000]
8/6	4/5	60209	[0.00000, 0.00000]	[0.00000, 0.00000]

Intervals use the fresh final-policy comparison stream, conditional on the original selection. Endpoints are rounded outward. Positive intervals identify higher native-ReLU cost; an interval containing zero is unresolved. Exact policy identities have an exact zero interval. There are twelve mathematical designs, each repeated on two workers with the same policies and stream, not twenty-four independent comparisons. N minus S is identically zero.

against exhaustive Fraction evaluation with the original smallest-index tie rule. An additional fixed subset receives seven shuffled repetitions per regime and worker. Coarser endpoint bounds and large-lattice guard cases are retained whether or not pruning is effective.

The eight methods separate symbolic reduction, vectorized enclosure, screening, exhaustive exact evaluation, native finite differences and adaptive subdivision. However, one direct contrast in that catalogue combines two changes: native exhaustive search uses the unreduced objective, whereas the native piecewise solver first reduces activation regions. We therefore execute a further two-by-two comparison before drawing an algorithm-specific timing conclusion. All four cells use the identical C++ arbitrary-size rational engine, objective, lattice, exact tie rule and

original-value postcheck. They cross activation reduction, off or on, with exhaustive or finite-difference search. Reduction is charged to both reduced implementations. The same ninety fixed rational instances, rather than an outcome-selected favorable subset, are used on two newly allocated worker jobs.

TABLE XVI
FINITE-DIFFERENCE VERSUS EXHAUSTIVE SEARCH AT THE SAME REDUCED REPRESENTATION

Width	Lattice	Faster 0	Faster 1	Saving 0 (%)	Saving 1 (%)
8	33	13/15	8/15	2.51	0.40
8	129	15/15	15/15	23.05	25.20
8	513	15/15	15/15	60.51	65.07
32	33	1/15	0/15	-17.79	-18.93
32	129	15/15	15/15	26.77	30.69
32	513	15/15	15/15	70.43	72.56

Both methods use the same C++ arbitrary-size rational engine, identical reduced objective and original-value postcheck. Each row includes three fixed seeds and all five regimes. Savings are medians of paired per-instance relative timings, not ratios of unrelated medians. Reduction, process creation, serialization and parsing are charged to both methods. The unreduced pair and all four-cell repeated-query results are retained in the supplement.

At a 513-point lattice, finite differences save a median of approximately 61–73 percent relative to native exhaustive search at the same reduced representation, depending on width and worker. Every instance in those cells is faster. At width 32 and lattice size 33, the median saving is negative on both workers, approximately –18 and –19 percent. Thus the operation-count advantage at large lattices has an observed small-lattice boundary; a root-free exact solver is not uniformly the cheapest implementation. The separate width-eight, 33-point cell is close to the process-overhead floor and is more sensitive to the worker and instance. The complete four-cell repetition table retains these effects rather than attributing a language change or activation simplification to screening alone.

Batching supplies another measured distinction. The complete ninety-query catalogue is evaluated at batch sizes one, eight and thirty-two for each native im-

plementation. Preprocessing, reduction where applicable, serialization, process creation, parsing and exact original-objective checks are all included. Every batch returns the same ninety exact answers. The batch study changes only the boundary of an implementation service; it does not create a new fitted policy or a new economic stopping experiment. Its full timings and byte counts are reported in the supplement, separately from the complete policy-service clocks.

Flatness and cancellation also clarify what a lossless result means. An interval screen is allowed to retain every minimizer and many nonminimizers; it need not produce one survivor. The finite-difference construction instead uses the exact polynomial piece, retaining the first minimizer even on a flat segment. Adaptive subdivision can reach its node budget and call an exact fallback. A fallback preserves correctness but charges its prefix work in addition to the subsequent solver. The full stress tables report all fallback events, endpoint resolutions, recorded operand lengths and interval-array memory. Endpoint coarsening after an exact rational bound is not described as a change in internal floating-point precision.

18.4. *Two controls and an additional innovation*

The multi-action calculation uses saved, genuinely trained two-state ReLU continuations at two specified states. The two controls share capacity $a_1 + a_2 \leq 1/8$. Their exposure columns are $(1/2, 1/4)$ and $(1/4, 1/2)$, and the optional second independent shock has a different exposure. Quantum denominators 64 and 128 give 45 and 153 feasible lattice points. Setting the second action to zero and disabling the second shock recovers the original action-dependent objective.

Both exact exhaustive evaluation and verified subdivision return the original constrained minimizer in every recorded query. The analytic uniform-box formula includes all shock dimensions actually used, with signed fitted output weights. Fewer exact point evaluations do not eliminate the cost of computing rectangle bounds: the adaptive/exhaustive time ratio is therefore reported along with evaluations and visited nodes. These experiments validate and cost a non-scalar witness extension. They do not constitute a complete two-control policy-cost or welfare

TABLE XVII
CONSTRAINED TWO-CONTROL EXACT WITNESS RECOVERY

Cap index	Shocks	Designs	Exhaustive	Adaptive	Max nodes	Fallback	Time ratio
8	1	4	45	7.0	107	0	3.24
8	2	4	45	7.0	107	0	3.19
16	1	4	153	80.0	256	4	2.70
16	2	4	153	80.0	256	4	2.68

Each of four mathematical designs per row uses a saved genuinely fitted ReLU continuation, two state points and two training seeds, repeated on two workers. Shared capacity is $1/8$, with quantum denominator eight times Cap index. Adaptive counts are selected exact evaluations; node-bound computation is additional work. Time ratio is the median paired adaptive/exhaustive query time over the eight records. A ratio above one means slower adaptive execution despite fewer exact evaluations.

These are action-search results, not complete two-control welfare comparisons.

study, and the scalar quartic search theorem is not invoked for a two-shock objective outside its stated degree structure.

18.5. Attaining an original Bellman-accuracy target

The direct Bellman calculation retains the original two-state, two-date economy. Lower intervals cover every continuous feasible action, whereas upper actions are quantum-valued and feasible throughout each acquired cell. Terminal innovation integration is analytic; unresolved earlier expectations use enclosures over the full original innovation bins. The uncentered procedure gives a valid but wide bracket. Centering at the same zero-policy continuation on both sides preserves cancellation before the gap is formed.

The centered ladder reaches the all-state tolerances $1/2$, $1/4$, $1/8$ and $1/16$. At $N = 128$, its maximum datewise gap is at most 0.03979, compared with approximately 0.30330 at the final uncentered rung. The centered construction does not attain $1/32$ on its declared ladder. Its cumulative recorded work includes every earlier centered rung and serialized endpoint array. These are constructive bounds against the original continuous-action optimum, not confidence intervals for gains relative to zero investment. The state partition here is a tensor verification grid;

TABLE XVIII

ORIGINAL-MODEL WHOLE-DOMAIN BELLMAN ACCURACY

N	A	q	Uncentered gap	Centered gap	Centered work (s)
8	8	4	2.18231	0.62981	0.478
16	16	8	1.12578	0.28126	2.290
32	32	16	0.59109	0.14409	11.288
64	64	32	0.30286	0.07289	80.081
128	64	32	–	0.03979	289.742

The model remains the original two-state, two-date investment economy with its continuous feasible actions and innovation law. N is the number of verification cells per state coordinate. The common-reference lower and upper endpoints cancel the same reference value pointwise. Centered work includes every preceding declared rung and endpoint serialization on that execution. The uncentered gap is the corresponding separately executed numerical bracket, not a same-host timing comparator. Tolerances $1/2$, $1/4$, $1/8$ and $1/16$ are attained; $1/32$ is not.

the non-tensor prospective actor and its high-dimensional service evidence remain distinct objects.

The bracket by itself certifies the policy constructed by that Bellman calculation. To assess the policies actually returned by the prospective services, we separately execute Proposition 26. The complete two-date service catalogue has 32 returned records and 5 distinct policy-plus-partition identities. Every identity is included, whether its policy came from a neural, quadratic or common-action generator. The lower table is fixed in advance at $N = 128$, $A = 64$, $q = 32$. The upper recursion is recomputed for each original actor through all intersecting acquired leaves. Neither training nor action selection is repeated.

Three of these returned actors have all-state bounds between approximately 0.06778 and 0.07176, and attain tolerance $1/8$ but not $1/16$. The other two retain bounds around 0.30330. The policy distinction is economically important: a service that meets a ten-percent initial-law improvement target can still return an actor with a materially wider worst-state optimality account. Conversely, the tighter fixed-policy bounds establish original-optimum accuracy for actual deployed actors rather than assigning the Bellman comparator's bound to an unrelated neural policy.

TABLE XIX

REVALIDATION OF EVERY DISTINCT ACTUALLY RETURNED TWO-DATE ACTOR

Actor	Leaves	Modes	Aliases	Date 0 gap	All-date gap	Unclipped	Own work (s)
2cc41f05	64	N/S	4	0.07176	0.07176	0.07176	2.254
3f2668fe	64	C/Q	8	0.07176	0.07176	0.07176	2.278
4e5b6e26	32	N/S	4	0.23801	0.30330	0.30330	2.228
718597a7	64	N/S	4	0.06778	0.06778	0.06778	2.279
a06e05ba	32	C/N/Q/S	12	0.23801	0.30330	0.30330	2.223

All thirty-two returned service records are represented by five complete policy-plus-partition identities.

No actor is retrained, improved or replaced. Verification uses $N=128$ and $q=32$ with the deposited centered lower Bellman table. Closed actor ranges include every intersecting leaf. The independent global nonworsening bound is intersected with upper excess endpoints, but the maximum gaps here coincide with the retained unclipped calculation. Own work covers each fixed-actor recursion through its endpoint archive; the shared lower-table reconstruction and process-return overhead are separate audit records.

The maximum gap is unchanged by the independently justified zero intersection in these results; the unclipped endpoints and gaps are also retained.

The shared lower-table reconstruction, each actor's additional upper recursion and the whole-process receipt are charged separately. These post-selection verification costs are not spliced into a fictitious earlier stopping run, and no old service clock is replaced by a replay timing. Deterministic complete-domain verification adds no independent policy-cost observation.

18.6. *Economic interpretation and computational scope*

For two implementations returning the same policy, the gain is a reduction in computation cost, conditional on the measured execution environment. The setup-and-use formula above gives the full range of implementation charges and use counts supported at any stated price of the work unit. For different policies, a returned-policy gain interval $[l, u]$, a replacement fee τ and an incremental measured-work

price $\lambda\Delta w$ give the net-benefit interval

$$[l - \tau - \lambda\Delta w, u - \tau - \lambda\Delta w].$$

A positive lower endpoint supports replacement under that price specification; a negative upper endpoint rules it out; the remaining cases are unresolved. This algebra does not estimate an installation fee, an electricity price or a welfare normalization from a hosted-runner clock.

The experiments specify CPU affinity and record the available processor, compiler and runtime metadata. Processor frequency is not controlled and hardware counters were unavailable under the recorded permissions. Separate worker allocations do not establish independent physical architectures. Rational operation counts, observed operand-size diagnostics, interval-array bytes, process resident memory, cold compilation and complete return times therefore remain separate quantities. The new evidence supports a terminating, costed exact witness and an executable original-optimum certificate within the original NBO development. It does not require suppressing adverse conventional comparisons or inferring an unexecuted scaling result.

19. PROSPECTIVE ACCURACY AND COMPLETE ECONOMIC POLICIES

The original-optimum question is now asked before execution, for every declared generator and economic cell. The previous service catalogue stopped on an initial-law gain relative to zero investment. That remains a meaningful adoption question, but it is not a certificate against the Bellman optimum. The present experiment makes the latter the common accuracy contract and separately evaluates the actual cost of each complete returned policy.

19.1. *Design and actual returned objects*

The catalogue has five cells: two-state economies with horizons two and three, a four-state economy with horizon three, an eight-state economy with horizon two, and the nested two-state, two-control economy with horizon two. The original scalar law and discount 15/16 are unchanged. The last cell implements both controls at

every date, with the capacity simplex, cross-action cost and additional continuous innovation stated in Section 12. It is not the earlier isolated two-control action query.

For each cell, the common comparator constructs nodal Bellman lower and upper bounds from primitive costs and transitions. Each fitted generator constructs its own future labels using 512 training states per date. The ReLU fit has 32 learned hidden units and 120 optimization iterations; the quadratic fit uses the same number of states. Four seeds, 6201–6204, are fixed before production. Each fitted continuation supplies proposals on the same 32-leaf state partition. The scalar proposal uses the retained exact root-free action recovery. The two-control proposal uses conventional sequential quadratic programming from four fixed starts, followed by downward lattice rounding and a feasibility check. Its approximate objective and termination flag are not treated as a certificate. The original-optimum verifier supplies the postcheck for both types of proposals.

The finest proposed state grids have $n = 64$ in two states, $n = 16$ in four states and $n = 4$ in eight states. Every preceding declared rung is executed and retained. Scalar and two-control action meshes have sixteen subdivisions, except the eight-state cell, which has eight. Every continuous innovation component has four bins, except that cell, which has two. No extra rung is added after a target fails. The five common tolerances are $1/2$, $1/4$, $1/8$, $1/16$ and $1/32$.

At each nodal state we preserve both the pure fitted proposal and the proposal augmented by the same common action family. The latter chooses the smaller verified original-continuation upper endpoint, retaining the common candidate on a tie. The nodal actions are then interpolated barycentrically, observed at forty bits and rounded downward at twenty action bits. These are explicitly new implementable policies. We do not call this operation revalidation of an unchanged discontinuous actor. The preceding fixed-policy study remains the place where the earlier actors are verified without changing them.

19.2. Matched all-state accuracy

Theorem 28 is applied to every actual returned actor. Its allowance includes the entire continuous-action optimum, off-grid state error, continuous innovation

integration, arithmetic, state acquisition, action rounding and all remaining dates. Neither an initial-state average nor a favorable cost difference can substitute for the maximum over states and dates in the stopping comparison.

TABLE XX

WHOLE-DOMAIN LOSS AGAINST THE ORIGINAL CONTINUOUS-ACTION OPTIMUM

$d/m/T$	n	Common	Pure ReLU	Augmented ReLU	Augmented quad.
2/1/2	64	0.00374	0.04924–0.08355	0.00374–0.00374	0.00374–0.00374
2/1/3	64	0.00759	0.09534–0.12293	0.00759–0.00759	0.00759–0.00759
4/1/3	16	0.05712	0.20915–0.22931	0.05712–0.05712	0.05712–0.05712
8/1/2	4	0.37961	0.51561–0.51923	0.37961–0.37961	0.37961–0.37961
2/2/2	64	0.00542	0.05710–0.08940	0.00542–0.00542	0.00542–0.00542

Each number bounds the maximum loss over all true states and all dates of the actual acquired, downward-rounded policy. Ranges are the minimum and maximum bound over all four fitting seeds, not confidence intervals. All displayed upper bounds are rounded upward. The common construction and every fitted actor use the same original-optimum reference.

The main table shows the final rung and the full range over the four seeds. The supplementary target and refinement tables give every target and every rung. In particular, the accuracy of the common comparator is not assigned to a pure neural policy. Pure policies have their own selected-action upper arrays and their own complete error recursions. Augmented policies receive a certificate for the actions they actually select, not a retrospective claim that the neural component alone produced the result.

The new second-order account differs from the previous range-table calculation in two ways. Positive interpolation preserves nodal information about the true convex Bellman function, rather than paying its full variation over every reached rectangle. Joint convexity also transfers nodal actions to an off-grid policy without imposing smoothness on a discontinuous actor. Both steps use the proved primitive regularity; neither is an empirical estimate of an unknown Bellman derivative. The retained tensor implementation exposes its dimension dependence, including the eight-state budget, rather than claiming a dimension-free solver.

19.3. *Direct policy costs and the adoption decision*

All final-rung actors are evaluated under the original uniform initial law and continuous innovation laws on 65,536 fresh common paths per cell. Forty-bit bins enclose each continuous state and innovation realization, and outward propagation encloses the cost of the implemented acquired policy. The path endpoints, not a fitted value at the initial state, enter the bounded empirical-Bernstein calculation. A single error budget of $1/100$ covers the prespecified finite family across all five cells, seeds and variants. Repeated seeds define different fitted objects; common paths induce dependence between their comparisons, which is allowed by the simultaneous coverage construction. Exact policy identities are reported as identities.

The sign convention is neural cost minus common cost. A wholly negative interval identifies a lower neural-policy cost. An interval containing zero does not establish equality or superiority. The supplement gives every marginal cost, every corresponding quadratic contrast and every zero-investment adoption gain; all four seeds are retained, not just the best-performing continuation. A change in the selected witness or in an all-state upper bound is therefore not used as a substitute for an economic comparison.

For an installation fee $\tau \geq 0$, let $[l, u]$ be the simultaneous interval for zero-policy cost minus the proposed controller's cost. Adoption has net gain in $[l - \tau, u - \tau]$. Every fee below a positive l is supported as beneficial; every fee above u is ruled out; the remaining range is unresolved. The same calculation applies to replacement of an installed pure fitted controller by its augmented controller using the separately recorded paired contrast. These are actual complete-policy decisions in the normalized economy. The intervals do not estimate an institutional installation charge, and a gain over zero investment does not establish a gain over the common Bellman comparator.

19.4. *Complete work at a common economic contract*

The cost of verification is part of the method. The reference lower and upper arrays are computed from primitives for every rung. The fitting/proposal service, actor transfer, every selected-action verification and serialization are then added.

TABLE XXI

DIRECT EXPECTED-COST CONTRASTS FOR EVERY NEURAL SEED

<i>d/m/T</i>	Seed	Pure ReLU minus common	Augmented ReLU minus common
2/1/2	6201	[0.00312, 0.01519]	[-0.00588, 0.00588]
2/1/2	6202	[0.00587, 0.01806]	[-0.00588, 0.00588]
2/1/2	6203	[0.00088, 0.01293]	[-0.00588, 0.00588]
2/1/2	6204	[0.00101, 0.01307]	[-0.00588, 0.00588]
2/1/3	6201	[0.01350, 0.02868]	[-0.00735, 0.00735]
2/1/3	6202	[0.00823, 0.02333]	[-0.00735, 0.00735]
2/1/3	6203	[0.00479, 0.01988]	[-0.00735, 0.00735]
2/1/3	6204	[0.01066, 0.02580]	[-0.00735, 0.00735]
4/1/3	6201	[0.02189, 0.03729]	[-0.00734, 0.00734]
4/1/3	6202	[0.02242, 0.03772]	[-0.00734, 0.00734]
4/1/3	6203	[0.01833, 0.03358]	[-0.00734, 0.00734]
4/1/3	6204	[0.02529, 0.04062]	[-0.00734, 0.00734]
8/1/2	6201	[0.01368, 0.02598]	[-0.00588, 0.00588]
8/1/2	6202	[0.01087, 0.02315]	[-0.00588, 0.00587]
8/1/2	6203	[0.01080, 0.02307]	[-0.00588, 0.00588]
8/1/2	6204	[0.01035, 0.02262]	[-0.00588, 0.00587]
2/2/2	6201	[0.00805, 0.02022]	[-0.00588, 0.00588]
2/2/2	6202	[0.00930, 0.02147]	[-0.00588, 0.00588]
2/2/2	6203	[0.00557, 0.01768]	[-0.00588, 0.00588]
2/2/2	6204	[0.00614, 0.01828]	[-0.00588, 0.00588]

Negative intervals identify lower neural-policy cost; intervals containing zero do not rank policies.

These are paired original-law cost intervals for complete final-rung actors. All five tasks and four seeds are retained on one simultaneous coverage event. Exact identity intervals are zero, not new independent observations. The supplement also reports every quadratic contrast and every marginal cost.

To avoid inventing unobserved marginal costs, each method is charged the entire shared reference and the entire shared inference service. Pure and augmented policies are verified together; each is conservatively charged that joint work. Cold compilation is included in the complete-release total and separately identified in its receipt.

TABLE XXII

RECORDED WORK THROUGH COMPLETE CATALOGUE RELEASE

$d/m/T$	Common (s)	ReLU (s)	Quadratic (s)	Reference (s)
2/1/2	14.14	14.69445–14.74057	14.38255–14.38457	2.41
2/1/3	17.80	18.77087–18.80325	18.16357–18.16561	3.68
4/1/3	70.46	75.06594–75.08827	74.36781–74.38030	51.71
8/1/2	106.06	118.18895–118.26983	117.75525–117.79752	84.37
2/2/2	94.31	99.70491–100.50574	98.92332–98.95254	80.91

Every method is charged the entire shared reference, all preceding rungs, its own fitting/proposal and verification work, full shared cost inference, and cold native compilation. Pure and augmented variants are jointly verified and each is conservatively charged the full joint work. This is measured catalogue-release work, not an unexecuted minimal stopping schedule. Times are one-host descriptive measurements; frequency was not controlled.

For each tolerance, a method is eligible precisely when its recorded actual-policy certificate attains that tolerance. Minimizing recorded complete-catalogue work among eligible methods defines the finite release frontier. This answers which executed complete catalogue supplies a certified controller at the stated contract. It does not estimate the minimum work of a sequential algorithm that was never run. The recorded first-crossing verification prefixes remain separate from this complete-release quantity, since the present independent cost service evaluates the final catalogue jointly.

The timing evidence is descriptive for the recorded host and arithmetic engine. Processor frequency was not controlled, and hardware counters were unavailable. The exact native witness retains its rational operand and request/response diagnostics. The new interpolation kernel reports explicit array sizes and arithmetic allowances, but logical input/output bytes are not presented as measured memory-bus traffic. No worker repetition is counted as an additional independent economic observation.

19.5. *Reliability and interpretation*

The four seeds define a finite fitting catalogue, not a probability model for optimizer success in an unspecified future economy. A pure fitted actor that fails an original-optimum target remains a failure in that catalogue even when its augmented counterpart succeeds. The count therefore separates the ability of the fit to supply a sufficiently accurate policy from the ability of the common verified fallback to complete the economic contract.

This execution advances the original NBO objective from a conditional action witness to a full original-optimum account across state dimensions, horizons and a genuinely implemented two-control policy. The preceding scalar exact-search result retains its stated function-class and lattice assumptions. The new policy theorem is independent of those assumptions once a feasible nodal proposal has been supplied. This extends certification to other proposal generators without asserting a new exact solver for deep networks or arbitrary shock distributions. Earlier conventional wins, failed targets, adverse multi-action query timings and broader economic applications remain available under their original hypotheses.

20. CONCLUSION

Neural Bellman Operators connect an own-future continuation to a feasible action and an economic cost account. The constructive backend retains the original Bellman-accuracy objective. The trained backend now has analytic innovation integration and a terminating exact action-lattice search, while its proposals remain subject to whole-cell true-advantage verification. These are complementary parts of the same numerical method, not interchangeable claims about the name of a function class.

The non-tensor cache and signed reference calculation turn the policy account into an executable prospective service. Their proofs retain the cost of leaf membership, critic evaluation, state dimension and horizon. The declared expected-cost targets are attained in the sensitivity experiment, with genuine nonterminal action changes and complete own-service clocks. Conventional generators share the same verification treatment, and policy identities make their role observable. The finite ex-

periments identify successful returns and their costs; they do not establish universal neural dominance or calibrated welfare effects.

The full-action gap, the actual expected cost and the price of acquiring or verifying a policy remain separate economic quantities. Keeping all three explicit permits a decision maker to choose the appropriate contract without treating fitting accuracy, certificate sharpness or a favorable timing observation as a substitute for the cost of the implemented policy.

The new action-search execution completes another part of that chain. Exact rational witnesses eliminate the stored critic's lattice optimization loss, and the transfer theorem identifies the remaining acquisition, evaluation, continuous-cover and verification obligations. The nested-candidate comparison makes a stricter certified upper endpoint attributable to an added witness under a fixed verifier. New prospective services and independent returned-policy contrasts determine the actual economic consequences. Neither exact fitted minimization nor a tighter upper endpoint is substituted for those contrasts. The complete records retain all modes, seeds, targets, attempted refinements and computational costs.

The lossless-search result makes that distinction operational at an unchanged answer. Exact activation-region reduction and outward screening preserve the trained-neural witness; conditional transcript invariance preserves the returned policy, all-state certificate and first-attainment rule. The separately frozen matched execution records lower complete-process median work in each declared task–target group, while retaining the unsuccessful target as unsuccessful. Its equal-policy resource comparison is distinct from the earlier comparisons between different neural and conventional policies. The original Bellman objective, feasibility requirements and economic laws are unchanged.

The additional exact construction makes the fitted action contract robust to flatness, adjacent ties and cancellation without assuming that an interval screen isolates one action. Its constrained-lattice extension separates action dimension from state dimension and gives an explicit integration and fallback account. The language- and representation-matched native factorial shows where the finite-difference algorithm reduces measured work and where a short lattice makes exhaustive evaluation

cheaper. Complete prospective services retain their own return boundary, conventional comparators and unsuccessful targets.

Common-reference Bellman verification further connects that computation to the original optimum. The constructed two-date policy attains an all-state one-sixteenth tolerance on the declared ladder. Revalidating every distinct actually returned two-date actor, without changing its parameters or actions, gives a separate accuracy account for deployed policies. Fixed-cover stability, full continuous-action lower covers and closed actor ranges are explicit conditions in that argument. The new certificates therefore strengthen the original NBO accuracy objective rather than replacing it with a gain relative to a convenient initial policy.

The original-optimum calculation now extends across the declared state dimensions and horizons and to a complete two-control economy. Its primitive regularity theorem closes a backward convexity and semiconcavity argument at the original discount. The policy certificate separately accounts for positive interpolation, continuous-action covering, original-law quadrature, acquisition, arithmetic and action rounding. These terms connect the actual returned controller to the Bellman optimum, while the prospective cost intervals determine the economic comparisons supported by the data.

The pure fitted and common-augmented variants answer different implementation questions. The latter may complete an accuracy contract that the former does not, and its full reference work must be charged. A changed witness or a lower certified endpoint is not by itself an identified cost advantage. The complete fixed catalogue retains these distinctions, including unsuccessful targets and unresolved rankings. No empirical calibration, controlled multi-architecture timing result or dimension-free computational guarantee is inferred from the executed study.

APPENDIX A: DEVELOPMENT AND REPRODUCIBILITY MAP

The complete development article and proof companion preserve every preceding theorem, economic application and numerical comparison, including adverse results. The prior active article and supplement are also retained as separate development documents. The present main article reorganizes the original constructive

core around trained action witnesses, non-tensor verification and prospective economic stopping; it does not replace the NBO subject with a different model.

The publication audit regenerates all unique recorded certificates from their saved fitted critics, replays all stopping decisions and verifies original moment bounds against exact rational sample moments. New solver regressions compare rational action witnesses with exhaustive lattice minimization. These operations do not create new policy-cost samples or replace historical service clocks. The source freezes, ordinary manuscript sources, complete service registry, raw arrays, response and release manifests are included in the revision directory. Mathematical peer review remains distinct from executable regression and document reproduction.

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